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Geometrical Optics

JEE Main Physics · Previous Year Questions with Answer Key

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Numerical	68 questions
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MCQ (Single Correct Answer) 217 questions

Q1. JEE Main 2026 (Online) 8th April Evening Shift

A thin biconvex lens is prepared from the glass $(\mu = 1.5)$ both curved surfaces of which have equal radii of 20 cm each. Left side surface of the lens is silvered from outside to make it reflecting. To have the position of image and object at the same place, the object should be placed, from the lens at a distance of ____ cm.

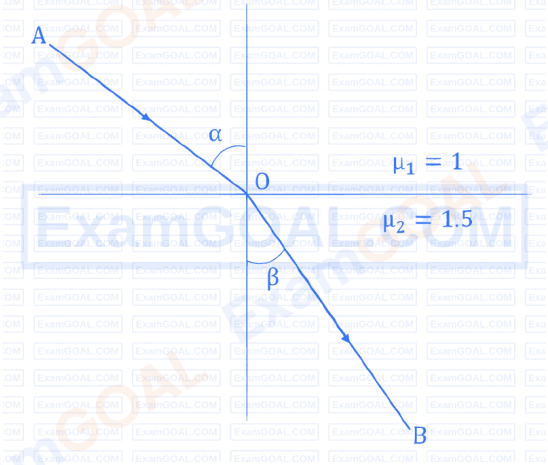
Open solution

(A) 10 (B) 12.5 (C) 13 (D) 13.5

Q2. JEE Main 2026 (Online) 8th April Evening Shift

Light ray incident along a vector $\overrightarrow{AO}(\overrightarrow{AO} = 2\hat{i} - 3\hat{j})$ emerges out along vector $\overrightarrow{OB}(\overrightarrow{OB} = C\hat{i} - 4\hat{j})$ as shown in the figure below. The value of C is ____ .

Open solution



(A) 1.6 (B) 0.16 (C) 11.6 (D) 16

Q3. JEE Main 2026 (Online) 6th April Evening Shift

Angle of minimum deviation is equal to the half of the angle of prism in an equilateral prism. The refractive index of the prism is ____

Open solution

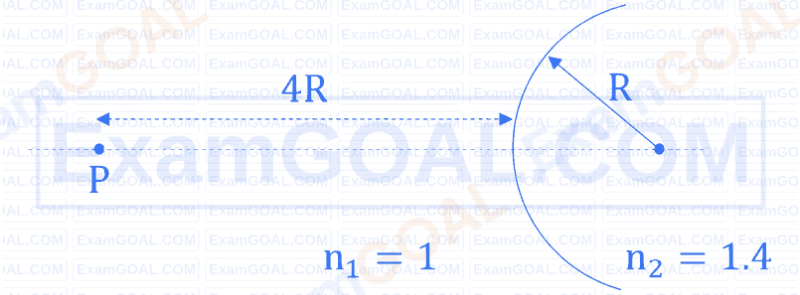
(A) 1.5 (B) $\sqrt{3}$ (C) $\sqrt{2}$ (D) 1.65

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Q4. JEE Main 2026 (Online) 6th April Morning Shift

A spherical interface lens of radius R separates two media of refractive indices 1 and 1.4 respectively as shown in the figure below. A point source is placed at a distance of $4R$ in front of spherical interface. The magnitude of the magnification of point source image is ____ .

Open solution



(A) 1.66 (B) 2.33 (C) 2.66 (D) 1.33

Q5. JEE Main 2026 (Online) 5th April Evening Shift

A thin convex lens and a thin concave lens are kept in contact and are co-axial. Which of the following statements is correct for this combination of two lenses ?

Open solution

(A) behaves as concave lens if $|f_{convex}| > |f_{concave}|$

(B) behaves as concave lens if $|f_{convex}| < |f_{concave}|$

(C) behaves as convex lens if $|f_{convex}| > |f_{concave}|$

(D) Focal length of the lens system will change if the positions of two lenses are interchanged

Q6. JEE Main 2026 (Online) 5th April Evening Shift

An object AB is placed 15 cm on the left of a convex lens P of focal length 10 cm . Another convex lens Q is now placed 15 cm right of lens P. If the focal length of lens Q is 15 cm , the final image is ____ .

Open solution

(A) virtual, formed at 7.5 cm right of lens Q, with a size bigger than that of AB

(B) real, formed at 7.5 cm right of lens Q, with a size same as that of AB

(C) formed at infinity.

(D) real, formed at 7 cm right of lens Q, with a size smaller than that of AB

Q7. JEE Main 2026 (Online) 5th April Morning Shift

A ray of light passing through an equilateral prism is having velocity $2.12 \times 10^8\ \mathrm{m/s}$ in the prism material, then the minimum angle of deviation is ____ degrees.

Open solution

(A) 45 (B) 30 (C) 28 (D) 58

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Q8. JEE Main 2026 (Online) 5th April Morning Shift

A compound microscope is designed with two symmetric biconvex lenses. The objective lens is cut vertically, creating two identical plano-convex lenses. One of them is used in place of original objective lens. To retain same magnification keeping the object distance unchanged, the tube length has to be

Open solution

(A) increased two times
(B) increased $\frac{3}{2}$ times
(C) decreased two times
(D) decreased $\frac{3}{2}$ times

Q9. JEE Main 2026 (Online) 4th April Evening Shift

A convex lens is made from glass material having refractive index of 1.4 with same radius of curvature on both sides. The ratio of its focal length and radius of curvature is ____ .

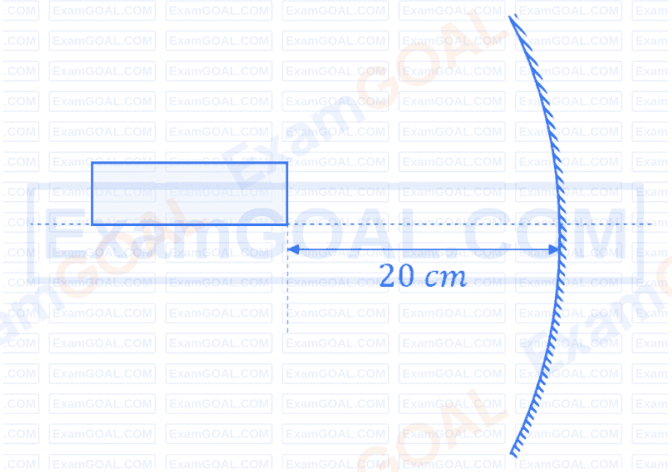
Open solution

(A) 0.5
(B) 2.5
(C) 0.8
(D) 1.25

Q10. JEE Main 2026 (Online) 4th April Morning Shift

A rod of length 10 cm lies along the principle axis of a concave mirror of focal length 10 cm as shown in figure. The length of the image is ____ cm.

Open solution



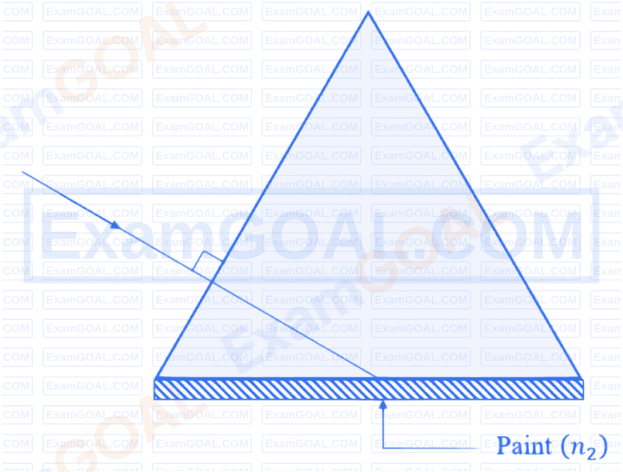
(A) 2.5
(B) 5
(C) 7.5
(D) 7

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Q11. JEE Main 2026 (Online) 2nd April Evening Shift

One side of an equilateral prism is painted by a transparent material of refractive index n_2. The refractive index of prism is 1.6. The minimum value of n_2 required for total internal reflection from painted face is _____.

Open solution



(A) $\frac{3\sqrt{3}}{1.6}$ (B) $\sqrt{3}$ (C) $\frac{3.2}{\sqrt{3}}$ (D) $\frac{4\sqrt{3}}{5}$

Q12. JEE Main 2026 (Online) 2nd April Morning Shift

For a thin symmetric prism made of glass (refractive index 1.5), the ratio of incident angle and minimum deviation will be _____.

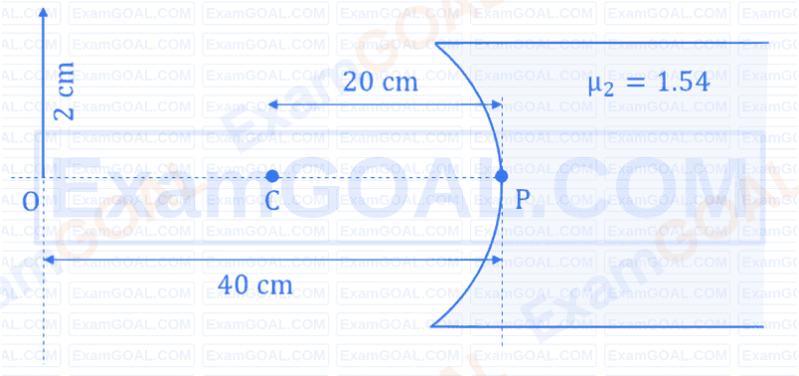
Open solution

(A) 3 : 4 (B) 3 : 2 (C) 2 : 1 (D) 1 : 2

Q13. JEE Main 2026 (Online) 2nd April Morning Shift

Refer the figure given below. μ_1 and μ_2 are refractive indices of air and lens material. The height of image will be _____ cm.

Open solution



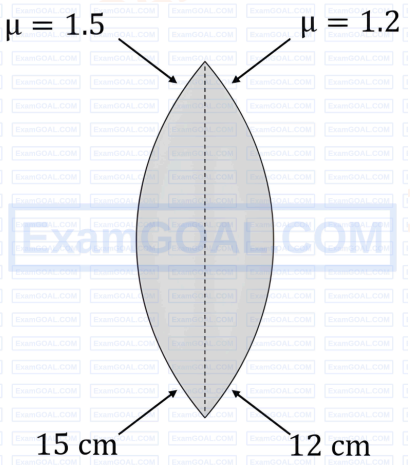
(A) 1 (B) 0.5 (C) 1.2 (D) 0.25

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Q14. JEE Main 2026 (Online) 28th January Evening Shift

Open solution

A biconvex lens is formed by using two thin planoconvex lenses, as shown in the figure. The refractive index and radius of curved surfaces are also mentioned in figure. When an object is placed on the left side of lens at a distance of 30 cm from the biconvex lens, the magnification of the image will be :



(A) −2 (B) −2.5 (C) +2.5 (D) +2

Q15. JEE Main 2026 (Online) 28th January Evening Shift

Open solution

For a transparent prism, if the angle of minimum deviation is equal to its refracting angle, the refractive index n of the prism satisfies.

(A) $n \geq 2$ (B) $\sqrt{2} < n < 2$ (C) $1 < n < 2$ (D) $\sqrt{2} < n < 2\sqrt{2}$

Q16. JEE Main 2026 (Online) 28th January Morning Shift

Open solution

The magnitudes of power of a biconvex lens (refractive index 1.5) and that of a plano-concave lens (refractive index $= 1.7$) are same. If the curvature of planoconcave lens exactly matches with the curvature of back surface of the biconvex lens, then ratio of radius of curvature of front and back surface of the biconvex lens is ____

(A) $2:5$ (B) $5:2$ (C) $12:5$ (D) $5:12$

Q17. JEE Main 2026 (Online) 24th January Evening Shift

Open solution

Distance between an object and three times magnified real image is 40 cm . The focal length of the mirror used is ____ cm .

(A) -20 (B) $-15/2$ (C) -10 (D) -15

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Q18. JEE Main 2026 (Online) 24th January Evening Shift

Five persons P_1, P_2, P_3, P_4 and P_5 recorded object distance (u) and image distance (v) using same convex lens having power +5 D as $(25, 96), (30, 62), (35, 37), (45, 35)$ and $(50, 32)$ respectively. Identify correct statement

Open solution

(A) Readings recorded by P_3 and P_2 persons are incorrect

(B) Readings recorded by all persons are correct

(C) Readings recorded by P_3 person are incorrect

(D) Readings recorded by P_4 and P_5 persons are incorrect

Q19. JEE Main 2026 (Online) 24th January Morning Shift

The exit surface of a prism with refractive index n is coated with a material having refractive index $\frac{n}{2}$. When this prism is set for minimum angle of deviation, it exactly meets the condition of critical angle. The prism angle is ____ .

Open solution

(A) 60° (B) 30° (C) 15° (D) 45°

Q20. JEE Main 2026 (Online) 24th January Morning Shift

An unpolarised light is incident at an interface of two dielectric media having refractive indices of 2 (incident medium) and $2\sqrt{3}$ (medium) respectively. To satisfy the condition that reflected and refracted rays are perpendicular to each other, the angle of incidence is ____

Open solution

(A) 45° (B) 60° (C) 10° (D) 30°

Q21. JEE Main 2026 (Online) 24th January Morning Shift

In a microscope of tube length 10 cm two convex lenses are arranged with focal length of 2 cm and 5 cm . Total magnification obtained with this system for normal adjustment is $(5)^k$. The value of k is ____ .

Open solution

(A) 4 (B) 5 (C) 2 (D) 3.5

Q22. JEE Main 2026 (Online) 23rd January Morning Shift

A thin prism with angle 5° of refractive index 1.72 is combined with another prism of refractive index 1.9 to produce dispersion without deviation. The angle of second prism is ____ .

Open solution

(A) 5° (B) 4° (C) 4.5° (D) 6°

Q23. JEE Main 2026 (Online) 23rd January Morning Shift

Consider light travelling from a medium A to medium B separated by a plane interface. If the light undergoes total internal reflection during its travel from medium A to B and the speed of light in media A and B are 2.4×10^8 m/s and 2.7×10^8 m/s, respectively, then the value of critical angle is :

Open solution

(A) $\cos^{-1}\left(\frac{8}{9}\right)$ (B) $\sin^{-1}\left(\frac{9}{8}\right)$

(C) $\cot^{-1}\left(\frac{3}{\sqrt{13}}\right)$ (D) $\tan^{-1}\left(\frac{8}{\sqrt{17}}\right)$

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Q24. JEE Main 2026 (Online) 22nd January Evening Shift

In parallax method for the determination of focal length of a concave mirror, the object should always be placed:

(A) between the focus (F) and the centre of curvature (C) of the mirror ONLY

(B) between the pole (P) and the focus (F) of the concave mirror ONLY

(C) at any point beyond the focus (F) of the mirror

(D) beyond the centre of the curvature (C) of the mirror ONLY

Open solution

Q25. JEE Main 2026 (Online) 22nd January Morning Shift

A thin convex lens of focal length 5 cm and a thin concave lens of focal length 4 cm are combined together (without any gap) and this combination has magnification m_1 when an object is placed 10 cm before the convex lens. Keeping the positions of convex lens and object undisturbed a gap of 1 cm is introduced between the lenses by moving the concave lens away, which lead to a change in magnification of total lens system to m_2. The value of $\left|\frac{m_1}{m_2}\right|$ is ____ .

(A) $\frac{25}{27}$ (B) $\frac{5}{6}$ (C) $\frac{5}{27}$ (D) $\frac{3}{2}$

Open solution

Q26. JEE Main 2026 (Online) 22nd January Morning Shift

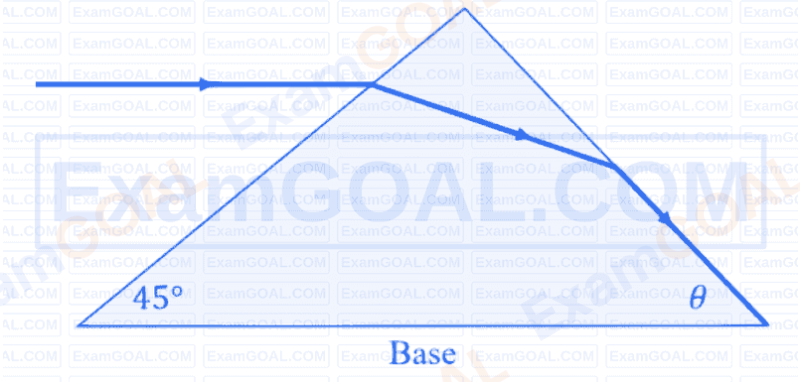
Consider an equilateral prism (refractive index $\sqrt{2}$). A ray of light is incident on its one surface at a certain angle i. If the emergent ray is found to graze along the other surface then the angle of refraction at the incident surface is close to ____

(A) 30° (B) 20° (C) 40° (D) 15°

Open solution

Q27. JEE Main 2026 (Online) 21st January Evening Shift

As shown in the diagram, when the incident ray is parallel to base of the prism, the emergent ray grazes along the second surface.



If refractive index of the material of prism is $\sqrt{2}$, the angle θ of prism is.

(A) 75° (B) 90° (C) 60° (D) 45°

Open solution

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Q28. JEE Main 2025 (Online) 8th April Evening Shift

A convex lens of focal length 30 cm is placed in contact with a concave lens of focal length 20 cm. An object is placed at 20 cm to the left of this lens system. The distance of the image from the lens in cm is _____.

Open solution

(A) $\frac{60}{7}$ (B) 15 (C) 45 (D) 30

Q29. JEE Main 2025 (Online) 8th April Evening Shift

A concave-convex lens of refractive index 1.5 and the radii of curvature of its surfaces are 30 cm and 20 cm, respectively. The concave surface is upwards and is filled with a liquid of refractive index 1.3. The focal length of the liquid-glass combination will be

Open solution

(A) $\frac{700}{11}$ cm (B) $\frac{600}{11}$ cm (C) $\frac{800}{11}$ cm (D) $\frac{500}{11}$ cm

Q30. JEE Main 2025 (Online) 7th April Evening Shift

Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A): Refractive index of glass is higher than that of air.

Reason (R): Optical density of a medium is directly proportionate to its mass density which results in a proportionate refractive index.

In the light of the above statements, choose the **most appropriate answer** from the options given below:

Open solution

(A) Both (A) and (R) are correct and (R) is the correct explanation of (A)

(B) (A) is correct but (R) is not correct

(C) (A) is not correct but (R) is correct

(D) Both (A) and (R) are correct but (R) is **not** the correct explanation of (A)

Q31. JEE Main 2025 (Online) 7th April Evening Shift

A mirror is used to produce an image with magnification of $\frac{1}{4}$. If the distance between object and its image is 40 cm, then the focal length of the mirror is _____.

Open solution

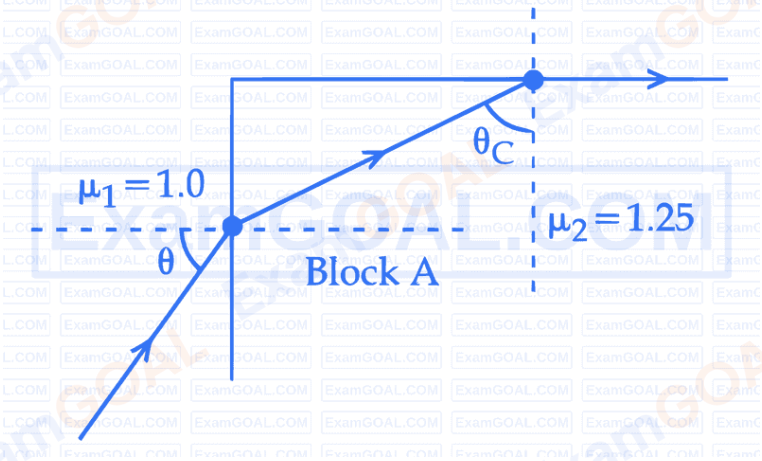
(A) 10 cm (B) 12.7 cm (C) 10.7 cm (D) 15 cm

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Q32. JEE Main 2025 (Online) 7th April Evening Shift

A transparent block A having refractive index $\mu = 1.25$ is surrounded by another medium of refractive index $\mu = 1.0$ as shown in the figure. A light ray is incident on the flat face of the block with incident angle θ as shown in the figure. What is the maximum value of θ for which light suffers total internal reflection at the top surface of the block?

Open solution



(A) $\tan^{-1}(4/3)$ (B) $\sin^{-1}(3/4)$ (C) $\tan^{-1}(3/4)$ (D) $\cos^{-1}(3/4)$

Q33. JEE Main 2025 (Online) 7th April Morning Shift

Two thin convex lenses of focal lengths 30 cm and 10 cm are placed coaxially, 10 cm apart. The power of this combination is:

(A) 5 D (B) 1 D (C) 20 D (D) 10 D

Open solution

Q34. JEE Main 2025 (Online) 7th April Morning Shift

A lens having refractive index 1.6 has focal length of 12 cm , when it is in air. Find the focal length of the lens when it is placed in water. (Take refractive index of water as 1.28)

(A) 355 mm (B) 655 mm (C) 288 mm (D) 555 mm

Open solution

Q35. JEE Main 2025 (Online) 4th April Evening Shift

A finite size object is placed normal to the principal axis at a distance of 30 cm from a convex mirror of focal length 30 cm . A plane mirror is now placed in such a way that the image produced by both the mirrors coincide with each other. The distance between the two mirrors is :

(A) 45 cm (B) 15 cm (C) 22.5 cm (D) 7.5 cm

Open solution

Q36. JEE Main 2025 (Online) 4th April Morning Shift

When an object is placed 40 cm away from a spherical mirror an image of magnification $\frac{1}{2}$ is produced. To obtain an image with magnification of $\frac{1}{3}$, the object is to be moved :

(A) 20 cm away from the mirror. (B) 20 cm towards the mirror.

(C) 80 cm away from the mirror. (D) 40 cm away from the mirror.

Open solution

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Q37. JEE Main 2025 (Online) 3rd April Morning Shift

Open solution

Consider following statements for refraction of light through prism, when angle of deviation is minimum.

A. The refracted ray inside prism becomes parallel to the base.

B. Larger angle prisms provide smaller angle of minimum deviation.

C. Angle of incidence and angle of emergence becomes equal.

D. There are always two sets of angle of incidence for which deviation will be same except at minimum deviation setting.

E. Angle of refraction becomes double of prism angle.

Choose the correct answer from the options given below :

(A) A, B and E Only (B) B, C and D Only (C) B, D and E Only (D) A, C and D Only

Q38. JEE Main 2025 (Online) 3rd April Morning Shift

Open solution

The radii of curvature for a thin convex lens are 10 cm and 15 cm respectively. The focal length of the lens is 12 cm . The refractive index of the lens material is

(A) 1.4 (B) 1.8 (C) 1.5 (D) 1.2

Q39. JEE Main 2025 (Online) 2nd April Evening Shift

Open solution

Two identical objects are placed in front of convex mirror and concave mirror having same radii of curvature of 12 cm , at same distance of 18 cm from the respective mirrors. The ratio of sizes of the images formed by convex mirror and by concave mirror is :

(A) 2 (B) $1/2$ (C) $1/3$ (D) 3

Q40. JEE Main 2025 (Online) 2nd April Evening Shift

Open solution

A bi-convex lens has radius of curvature of both the surfaces same as $1/6$ cm. If this lens is required to be replaced by another convex lens having different radii of curvatures on both sides $(R_1 \neq R_2)$, without any change in lens power then possible combination of R_1 and R_2 is :

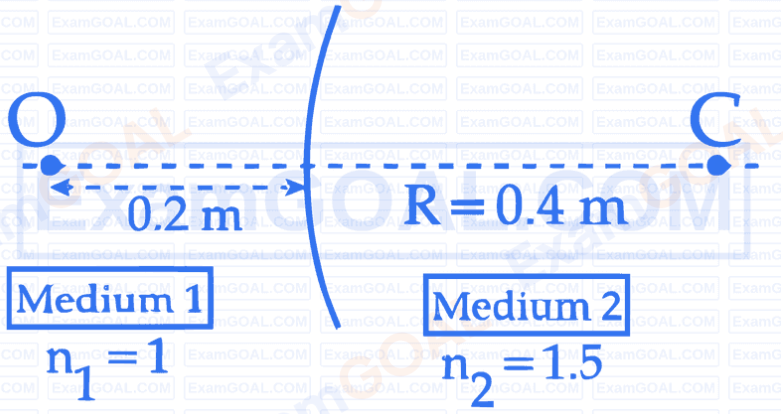
(A) $\frac{1}{3}$ cm and $\frac{1}{7}$ cm (B) $\frac{1}{5}$ cm and $\frac{1}{7}$ cm

(C) $\frac{1}{3}$ cm and $\frac{1}{3}$ cm (D) $\frac{1}{6}$ cm and $\frac{1}{9}$ cm

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Q41. JEE Main 2025 (Online) 2nd April Morning Shift

Open solution



A spherical surface separates two media of refractive indices 1 and 1.5 as shown in figure. Distance of the image of an object ' O ', is :

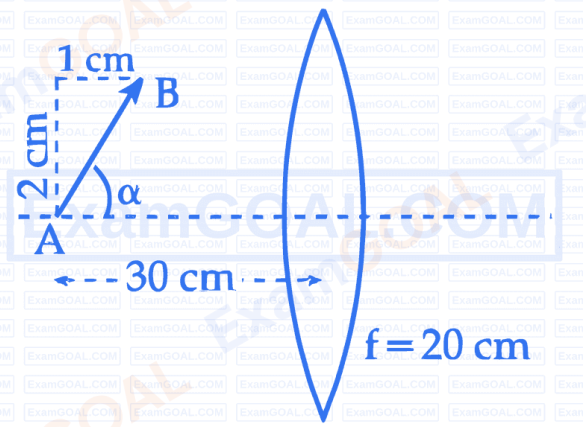
(C is the center of curvature of the spherical surface and R is the radius of curvature)

(A) 0.24 m left to the spherical surface
(B) 0.24 m right to the spherical surface
(C) 0.4 m left to the spherical surface
(D) 0.4 m right to the spherical surface

Q42. JEE Main 2025 (Online) 2nd April Morning Shift

Open solution

A slanted object AB is placed on one side of convex lens as shown in the diagram. The image is formed on the opposite side. Angle made by the image with principal axis is :



(A) $-\alpha$
(B) $-\frac{\alpha}{2}$
(C) -45°
(D) $+45^\circ$

Q43. JEE Main 2025 (Online) 29th January Evening Shift

Open solution

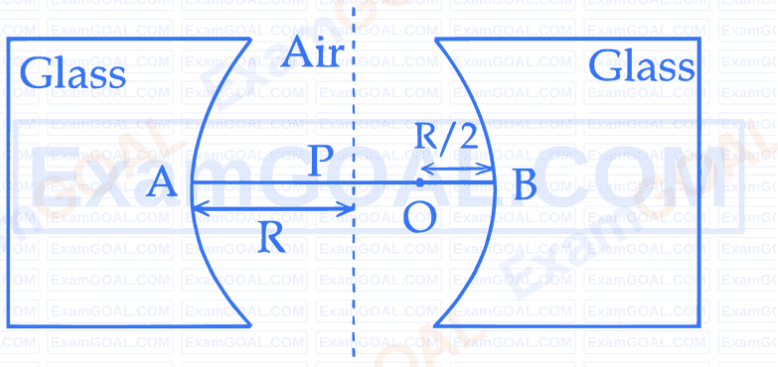
A convex lens made of glass (refractive index = 1.5) has focal length 24 cm in air. When it is totally immersed in water (refractive index = 1.33), its focal length changes to

(A) 96 cm
(B) 72 cm
(C) 24 cm
(D) 48 cm

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Q44. JEE Main 2025 (Online) 29th January Evening Shift

Open solution



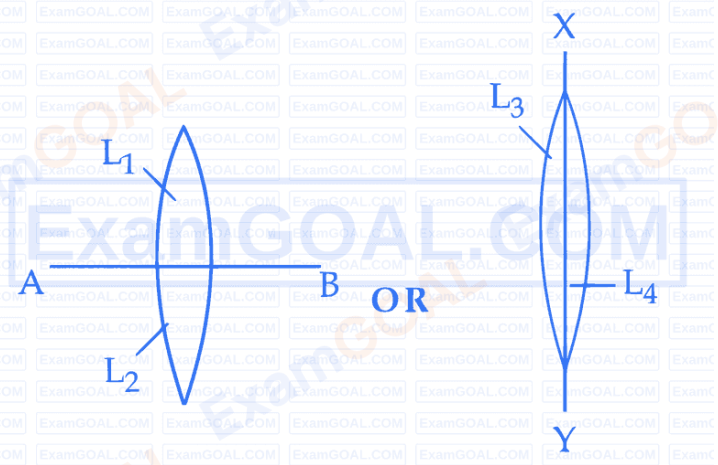
Two concave refracting surfaces of equal radii of curvature and refractive index 1.5 face each other in air as shown in figure. A point object O is placed midway, between P and B. The separation between the images of O, formed by each refracting surface is :

(A) 0.124R (B) 0.114R (C) 0.411R (D) 0.214R

Q45. JEE Main 2025 (Online) 29th January Evening Shift

Open solution

Two identical symmetric double convex lenses of focal length f are cut into two equal parts L_1, L_2 by AB plane and L_3, L_4 by XY plane as shown in figure respectively. The ratio of focal lengths of lenses L_1 and L_3 is



(A) 1 : 2 (B) 1 : 1 (C) 2 : 1 (D) 1 : 4

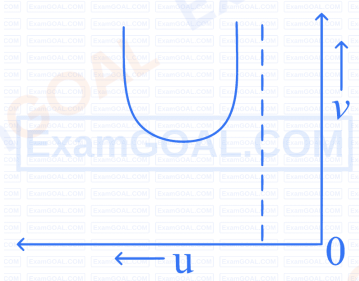
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Q46. JEE Main 2025 (Online) 29th January Morning Shift

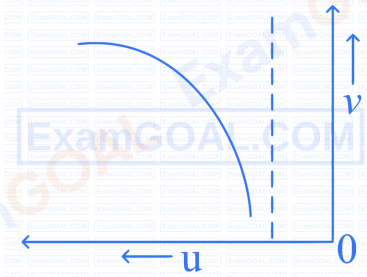
Open solution

Let u and v be the distances of the object and the image from a lens of focal length f. The correct graphical representation of u and v for a convex lens when $|u| > f$, is

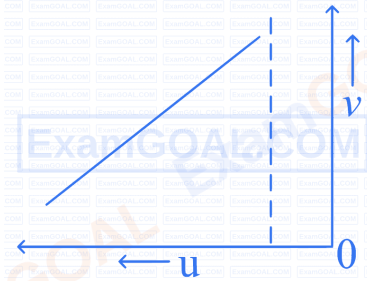
(A)



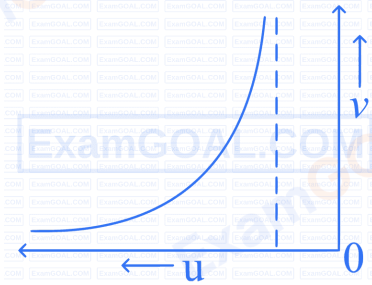
(B)



(C)



(D)



Q47. JEE Main 2025 (Online) 29th January Morning Shift

Open solution

At the interface between two materials having refractive indices n_1 and n_2, the critical angle for reflection of an em wave is θ_{1C}. The n_2 material is replaced by another material having refractive index n_3 such that the critical angle at the interface between n_1 and n_3 materials is θ_{2C}. If $n_3 > n_2 > n_1$; $\frac{n_2}{n_3} = \frac{2}{5}$ and $\sin\theta_{2C} - \sin\theta_{1C} = \frac{1}{2}$, then θ_{1C} is :

(A) $\sin^{-1}\left(\frac{1}{6n_1}\right)$

(B) $\sin^{-1}\left(\frac{1}{3n_1}\right)$

(C) $\sin^{-1}\left(\frac{5}{6n_1}\right)$

(D) $\sin^{-1}\left(\frac{2}{3n_1}\right)$

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Q48. JEE Main 2025 (Online) 28th January Evening Shift

A concave mirror produces an image of an object such that the distance between the object and image is 20 cm. If the magnification of the image is −3, then the magnitude of the radius of curvature of the mirror is :

Open solution

(A) 3.75 cm (B) 15 cm (C) 7.5 cm (D) 30 cm

Q49. JEE Main 2025 (Online) 28th January Evening Shift

In a long glass tube, mixture of two liquids A and B with refractive indices 1.3 and 1.4 respectively, forms a convex refractive meniscus towards A . If an object placed at 13 cm from the vertex of the meniscus in A forms an image with a magnification of -2, then the radius of curvature of meniscus is :

Open solution

(A) $\frac{2}{3}$ cm (B) $\frac{4}{3}$ cm (C) $\frac{1}{3}$ cm (D) 1 cm

Q50. JEE Main 2025 (Online) 28th January Morning Shift

A thin prism P_1 with angle 4° made of glass having refractive index 1.54 , is combined with another thin prism P_2 made of glass having refractive index 1.72 to get dispersion without deviation. The angle of the prism P_2 in degrees is

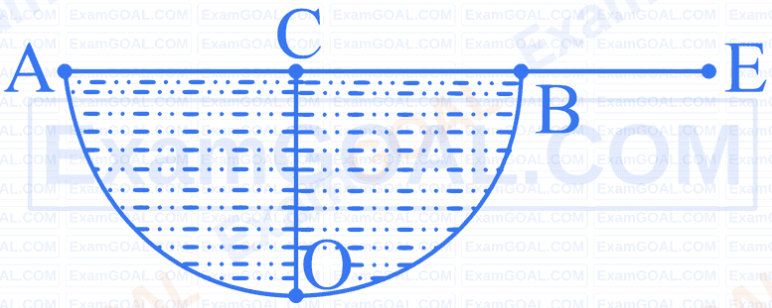
Open solution

(A) 1.5 (B) 16/3 (C) 3 (D) 4

Q51. JEE Main 2025 (Online) 28th January Morning Shift

A hemispherical vessel is completely filled with a liquid of refractive index μ. A small coin is kept at the lowest point (O) of the vessel as shown in figure. The minimum value of the refractive index of the liquid so that a person can see the coin from point E (at the level of the vessel) is _____.

Open solution



(A) $\frac{3}{2}$ (B) $\frac{\sqrt{3}}{2}$ (C) $\sqrt{3}$ (D) $\sqrt{2}$

Q52. JEE Main 2025 (Online) 24th January Evening Shift

A photograph of a landscape is captured by a drone camera at a height of 18 km . The size of the camera film is $2\ \text{cm} \times 2\ \text{cm}$ and the area of the landscape photographed is $400\ \text{km}^2$. The focal length of the lens in the drone camera is :

Open solution

(A) 2.8 cm (B) 0.9 cm (C) 2.5 cm (D) 1.8 cm

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Q53. JEE Main 2025 (Online) 24th January Morning Shift

What is the relative decrease in focal length of a lens for an increase in optical power by 0.1 D from 2.5D ? ['D' stands for dioptre]

(A) 0.04 (B) 0.1 (C) 0.40 (D) 0.01

Open solution

Q54. JEE Main 2025 (Online) 24th January Morning Shift

A thin plano convex lens made of glass of refractive index 1.5 is immersed in a liquid of refractive index 1.2. When the plane side of the lens is silver coated for complete reflection, the lens immersed in the liquid behaves like a concave mirror of focal length 0.2 m . The radius of curvature of the curved surface of the lens is

(A) 0.15 m (B) 0.10 m (C) 0.25 m (D) 0.20 m

Open solution

Q55. JEE Main 2025 (Online) 24th January Morning Shift

A plano-convex lens having radius of curvature of first surface 2 cm exhibits focal length of f_1 in air. Another plano-convex lens with first surface radius of curvature 3 cm has focal length of f_2 when it is immersed in a liquid of refractive index 1.2. If both the lenses are made of same glass of refractive index 1.5 , the ratio of f_1 and f_2 will be

(A) $2:3$ (B) $1:3$ (C) $1:2$ (D) $3:5$

Open solution

Q56. JEE Main 2025 (Online) 23rd January Evening Shift

A concave mirror of focal length f in air is dipped in a liquid of refractive index μ. Its focal length in the liquid will be:

(A) μf (B) f (C) $\frac{f}{\mu}$ (D) $\frac{f}{(\mu-1)}$

Open solution

Q57. JEE Main 2025 (Online) 23rd January Evening Shift

The refractive index of the material of a glass prism is $\sqrt{3}$. The angle of minimum deviation is equal to the angle of the prism. What is the angle of the prism?

(A) 60° (B) 50° (C) 58° (D) 48°

Open solution

Q58. JEE Main 2025 (Online) 23rd January Morning Shift

What is the lateral shift of a ray refracted through a parallel-sided glass slab of thickness ' h ' in terms of the angle of incidence ' i ' and angle of refraction ' r ', if the glass slab is placed in air medium?

(A) h (B) $\frac{h\cos(i-r)}{\sin r}$

(C) $\frac{\mathrm{h}\tan(\mathrm{i}-\mathrm{r})}{\tan \mathrm{r}}$ (D) $\frac{h\sin(i-r)}{\cos r}$

Open solution

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Q59. JEE Main 2025 (Online) 23rd January Morning Shift

Open solution

A spherical surface of radius of curvature R, separates air from glass (refractive index $= 1.5$). The centre of curvature is in the glass medium. A point object ' O ' placed in air on the optic axis of the surface, so that its real image is formed at 'I' inside glass. The line OI intersects the spherical surface at P and $PO = PI$. The distance PO equals to

(A) 5R (B) 2R (C) 1.5R (D) 3R

Q60. JEE Main 2025 (Online) 23rd January Morning Shift

Open solution

Given a thin convex lens (refractive index μ_2), kept in a liquid (refractive index $\mu_1, \mu_1 < \mu_2$) having radii of curvatures $|R_1|$ and $|R_2|$. Its second surface is silver polished. Where should an object be placed on the optic axis so that a real and inverted image is formed at the same place?

(A) $\frac{(\mu_2+\mu_1)|R_1|}{(\mu_2-\mu_1)}$

(B) $\frac{\mu_1|R_1|\cdot|R_2|}{\mu_2(2|R_1|+|R_2|)-\mu_1\sqrt{|R_1|\cdot|R_2|}}$

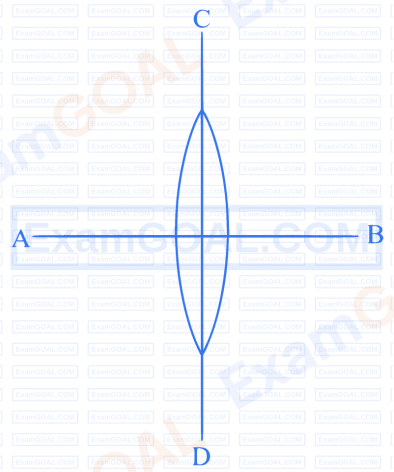
(C) $\frac{\mu_1|R_1|\cdot|R_2|}{\mu_2(|R_1|+|R_2|)-\mu_1|R_1|}$

(D) $\frac{\mu_1|R_1|\cdot|R_2|}{\mu_2(|R_1|+|R_2|)-\mu_1|R_2|}$

Q61. JEE Main 2025 (Online) 22nd January Evening Shift

Open solution

A symmetric thin biconvex lens is cut into four equal parts by two planes AB and CD as shown in figure. If the power of original lens is 4D then the power of a part of the divided lens is



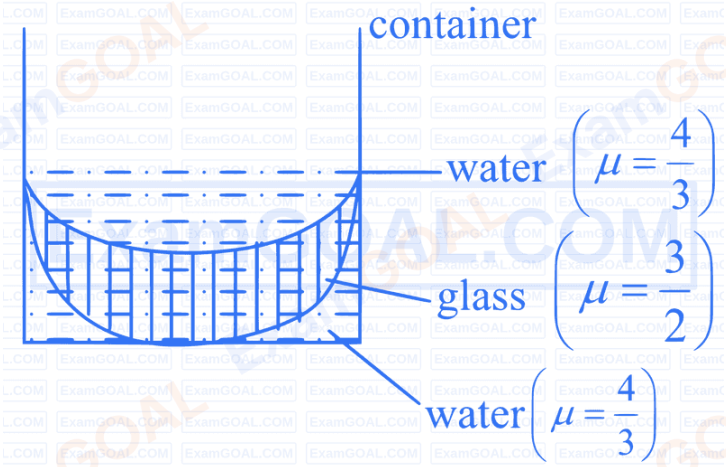
(A) 2D (B) 8D (C) 4D (D) D

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Q62. JEE Main 2025 (Online) 22nd January Morning Shift

Open solution

In the diagram given below, there are three lenses formed. Considering negligible thickness of each of them as compared to $|R_1|$ and $|R_2|$, i.e., the radii of curvature for upper and lower surfaces of the glass lens, the power of the combination is



(A) $-\frac{1}{6}\left(\frac{1}{|R_1|} - \frac{1}{|R_2|}\right)$

(B) $-\frac{1}{6}\left(\frac{1}{|R_1|} + \frac{1}{|R_2|}\right)$

(C) $\frac{1}{6}\left(\frac{1}{|R_1|} + \frac{1}{|R_2|}\right)$

(D) $\frac{1}{6}\left(\frac{1}{|R_1|} - \frac{1}{|R_2|}\right)$

Q63. JEE Main 2025 (Online) 22nd January Morning Shift

Open solution

Given is a thin convex lens of glass (refractive index μ) and each side having radius of curvature R. One side is polished for complete reflection. At what distance from the lens, an object be placed on the optic axis so that the image gets formed on the object itself?

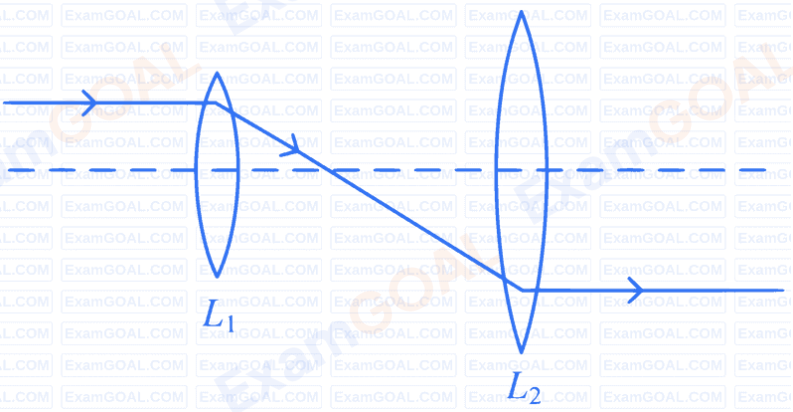
(A) R/μ (B) $R/(2\mu - 1)$ (C) $R/(2\mu - 3)$ (D) μR

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Q64. JEE Main 2024 (Online) 9th April Evening Shift

Open solution

The following figure represents two biconvex lenses L_1 and L_2 having focal length $10\ \text{cm}$ and $15\ \text{cm}$ respectively. The distance between L_1 & L_2 is :



(A) 25 cm (B) 35 cm (C) 10 cm (D) 15 cm

Q65. JEE Main 2024 (Online) 9th April Morning Shift

Open solution

Given below are two statements :

Statement (I) : When an object is placed at the centre of curvature of a concave lens, image is formed at the centre of curvature of the lens on the other side.

Statement (II) : Concave lens always forms a virtual and erect image.

In the light of the above statements, choose the correct answer from the options given below :

(A) Statement I is false but Statement II is true

(B) Both Statement I and Statement II are false

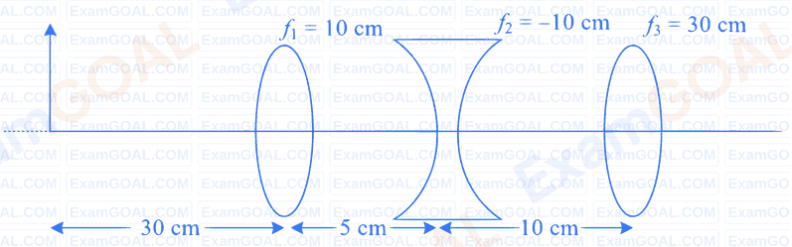
(C) Both Statement I and Statement II are true

(D) Statement I is true but Statement II is false

Q66. JEE Main 2024 (Online) 8th April Evening Shift

Open solution

The position of the image formed by the combination of lenses is :



(A) $15\ \text{cm}$ (right of second lens)

(B) $30\ \text{cm}$ (right of third lens)

(C) $30\ \text{cm}$ (left of third lens)

(D) $15\ \text{cm}$ (left of second lens)

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Q67. JEE Main 2024 (Online) 8th April Morning Shift

Critical angle of incidence for a pair of optical media is 45°. The refractive indices of first and second media are in the ratio:

(A) $1:2$ (B) $1:\sqrt{2}$ (C) $2:1$ (D) $\sqrt{2}:1$

Open solution

Q68. JEE Main 2024 (Online) 6th April Evening Shift

For the thin convex lens, the radii of curvature are at $15\ \text{cm}$ and $30\ \text{cm}$ respectively. The focal length of the lens is $20\ \text{cm}$. The refractive index of the material is :

(A) 1.2 (B) 1.5 (C) 1.4 (D) 1.8

Open solution

Q69. JEE Main 2024 (Online) 5th April Evening Shift

Given below are two statements :

Statement I : When the white light passed through a prism, the red light bends lesser than yellow and violet.

Statement II : The refractive indices are different for different wavelengths in dispersive medium. In the light of the above statements, choose the correct answer from the options given below :

(A) Both Statement I and Statement II are true

(B) Statement I is true but Statement II is false

(C) Statement I is false but Statement II is true

(D) Both Statement I and Statement II are false

Open solution

Q70. JEE Main 2024 (Online) 4th April Morning Shift

An effective power of a combination of 5 identical convex lenses which are kept in contact along the principal axis is 25D. Focal length of each of the convex lens is:

(A) 50 cm (B) 20 cm (C) 25 cm (D) 500 cm

Open solution

Q71. JEE Main 2024 (Online) 31st January Morning Shift

The refractive index of a prism with apex angle A is $\cot A/2$. The angle of minimum deviation is :

(A) $\delta_m = 180^\circ - 3\ \text{A}$ (B) $\delta_m = 180^\circ - 4A$

(C) $\delta_m = 180^\circ - 2A$ (D) $\delta_m = 180^\circ - A$

Open solution

Q72. JEE Main 2024 (Online) 29th January Evening Shift

If the distance between object and its two times magnified virtual image produced by a curved mirror is $15\ \text{cm}$, the focal length of the mirror must be:

(A) -10 cm (B) -12 cm (C) 15 cm (D) 10/3 cm

Open solution

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Q73. JEE Main 2024 (Online) 29th January Morning Shift

A convex mirror of radius of curvature $30\ \text{cm}$ forms an image that is half the size of the object. The object distance is :

(A) -45 cm (B) -15 cm (C) 45 cm (D) 15 cm

Open solution

Q74. JEE Main 2024 (Online) 29th January Morning Shift

A biconvex lens of refractive index 1.5 has a focal length of $20\ \text{cm}$ in air. Its focal length when immersed in a liquid of refractive index 1.6 will be:

(A) -16 cm (B) +16 cm (C) +160 cm (D) -160 cm

Open solution

Q75. JEE Main 2024 (Online) 27th January Morning Shift

If the refractive index of the material of a prism is $\cot\left(\frac{A}{2}\right)$, where A is the angle of prism then the angle of minimum deviation will be

(A) $\pi - 2\,\text{A}$ (B) $\frac{\pi}{2} - 2\,\text{A}$ (C) $\pi - \text{A}$ (D) $\frac{\pi}{2} - \text{A}$

Open solution

Q76. JEE Main 2024 (Online) 27th January Morning Shift

A convex lens of focal length $40\ \text{cm}$ forms an image of an extended source of light on a photoelectric cell. A current I is produced. The lens is replaced by another convex lens having the same diameter but focal length $20\ \text{cm}$. The photoelectric current now is :

(A) $\frac{\text{I}}{2}$ (B) 4 I (C) 2 I (D) I

Open solution

Q77. JEE Main 2023 (Online) 13th April Morning Shift

A vessel of depth 'd' is half filled with oil of refractive index n_1 and the other half is filled with water of refractive index n_2. The apparent depth of this vessel when viewed from above will be-

(A) $\frac{2d(n_1+n_2)}{n_1n_2}$ (B) $\frac{d(n_1+n_2)}{2n_1n_2}$

(C) $\frac{dn_1n_2}{2(n_1+n_2)}$ (D) $\frac{dn_1n_2}{(n_1+n_2)}$

Open solution

Q78. JEE Main 2023 (Online) 12th April Morning Shift

An ice cube has a bubble inside. When viewed from one side the apparent distance of the bubble is $12\ \text{cm}$. When viewed from the opposite side, the apparent distance of the bubble is observed as $4\ \text{cm}$. If the side of the ice cube is $24\ \text{cm}$, the refractive index of the ice cube is

(A) $\frac{3}{2}$ (B) $\frac{4}{3}$ (C) $\frac{2}{3}$ (D) $\frac{6}{5}$

Open solution

Q79. JEE Main 2023 (Online) 11th April Evening Shift

When one light ray is reflected from a plane mirror with 30° angle of reflection, the angle of deviation of the ray after reflection is :

(A) 140° (B) 130° (C) 120° (D) 110°

Open solution

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Q80. JEE Main 2023 (Online) 11th April Morning Shift

The critical angle for a denser-rarer interface is 45°. The speed of light in rarer medium is $3 \times 10^8\ \mathrm{m/s}$. The speed of light in the denser medium is:

(A) $2.12 \times 10^8\ \mathrm{m/s}$
(B) $5 \times 10^7\ \mathrm{m/s}$
(C) $\sqrt{2} \times 10^8\ \mathrm{m/s}$
(D) $3.12 \times 10^7\ \mathrm{m/s}$

Open solution

Q81. JEE Main 2023 (Online) 10th April Morning Shift

An object is placed at a distance of 12 cm in front of a plane mirror. The virtual and erect image is formed by the mirror. Now the mirror is moved by 4 cm towards the stationary object. The distance by which the position of image would be shifted, will be

(A) 4 cm towards mirror
(B) 2 cm towards mirror
(C) 8 cm away from mirror
(D) 8 cm towards mirror

Open solution

Q82. JEE Main 2023 (Online) 8th April Morning Shift

In a reflecting telescope, a secondary mirror is used to:

(A) make chromatic aberration zero
(B) remove spherical aberration
(C) reduce the problem of mechanical support
(D) move the eyepiece outside the telescopic tube

Open solution

Q83. JEE Main 2023 (Online) 6th April Evening Shift

Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R

Assertion A: The phase difference of two light waves changes if they travel through different media having same thickness, but different indices of refraction.

Reason R: The wavelengths of waves are different in different media.

In the light of the above statements, choose the most appropriate answer from the options given below

(A) Both A and R are correct but R is NOT the correct explanation of A
(B) A is correct but R is not correct
(C) A is not correct but R is correct
(D) Both A and R are correct and R is the correct explanation of A

Open solution

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Q84. JEE Main 2023 (Online) 6th April Evening Shift

A 2 meter long scale with least count of $0.2\ \text{cm}$ is used to measure the locations of objects on an optical bench. While measuring the focal length of a convex lens, the object pin and the convex lens are placed at $80\ \text{cm}$ mark and $1\ \text{m}$ mark, respectively. The image of the object pin on the other side of lens coincides with image pin that is kept at $180\ \text{cm}$ mark. The $\%$ error in the estimation of focal length is:

Open solution

(A) 1.70 (B) 0.51 (C) 1.02 (D) 0.85

Q85. JEE Main 2023 (Online) 6th April Morning Shift

A monochromatic light wave with wavelength λ_1 and frequency v_1 in air enters another medium. If the angle of incidence and angle of refraction at the interface are 45° and 30° respectively, then the wavelength λ_2 and frequency v_2 of the refracted wave are:

Open solution

(A) $\lambda_2 = \lambda_1, v_2 = \frac{1}{\sqrt{2}}v_1$ (B) $\lambda_2 = \lambda_1, v_2 = \sqrt{2}v_1$

(C) $\lambda_2 = \sqrt{2}\lambda_1, v_2 = v_1$ (D) $\lambda_2 = \frac{1}{\sqrt{2}}\lambda_1, v_2 = v_1$

Q86. JEE Main 2023 (Online) 1st February Evening Shift

Two objects A and B are placed at 15 cm and 25 cm from the pole in front of a concave mirror having radius of curvature 40 cm. The distance between images formed by the mirror is _____.

Open solution

(A) 60 cm (B) 40 cm (C) 160 cm (D) 100 cm

Q87. JEE Main 2023 (Online) 31st January Evening Shift

A microscope is focused on an object at the bottom of a bucket. If liquid with refractive index $\frac{5}{3}$ is poured inside the bucket, then the microscope has to be raised by $30\ \text{cm}$ to focus the object again. The height of the liquid in the bucket is :

Open solution

(A) $50\ \text{cm}$ (B) $18\ \text{cm}$ (C) $75\ \text{cm}$ (D) $12\ \text{cm}$

Q88. JEE Main 2023 (Online) 30th January Evening Shift

A thin prism P_1 with an angle 6° and made of glass of refractive index 1.54 is combined with another prism P_2 made from glass of refractive index 1.72 to produce dispersion without average deviation. The angle of prism P_2 is

Open solution

(A) 4.5° (B) 7.8° (C) 1.3° (D) 6°

Q89. JEE Main 2023 (Online) 30th January Morning Shift

A person has been using spectacles of power -1.0 dioptre for distant vision and a separate reading glass of power 2.0 dioptres. What is the least distance of distinct vision for this person :

Open solution

(A) 50 cm (B) 40 cm (C) 30 cm (D) 10 cm

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Q90. JEE Main 2023 (Online) 25th January Evening Shift

The light rays from an object have been reflected towards an observer from a standard flat mirror, the image observed by the observer is :-

A. Real

B. Erect

C. Smaller in size than object

D. Laterally inverted

Choose the most appropriate answer from the options given below :

(A) B and D only (B) A, C, and D only (C) A and D only (D) B and C only

Open solution

Q91. JEE Main 2023 (Online) 24th January Evening Shift

When a beam of white light is allowed to pass through convex lens parallel to principal axis, the different colours of light converge at different points on the principal axis after refraction. This is called :

(A) Spherical aberration (B) Scattering

(C) Polarisation (D) Chromatic aberration

Open solution

Q92. JEE Main 2022 (Online) 29th July Evening Shift

Light enters from air into a given medium at an angle of 45° with interface of the air-medium surface. After refraction, the light ray is deviated through an angle of 15° from its original direction. The refractive index of the medium is:

(A) 1.732 (B) 1.333 (C) 1.414 (D) 2.732

Open solution

Q93. JEE Main 2022 (Online) 28th July Evening Shift

The power of a lens (biconvex) is $1.25\ \mathrm{m}^{-1}$ in particular medium. Refractive index of the lens is 1.5 and radii of curvature are $20\ \mathrm{cm}$ and $40\ \mathrm{cm}$ respectively. The refractive index of surrounding medium:

(A) 1.0 (B) $\frac{9}{7}$ (C) $\frac{3}{2}$ (D) $\frac{4}{3}$

Open solution

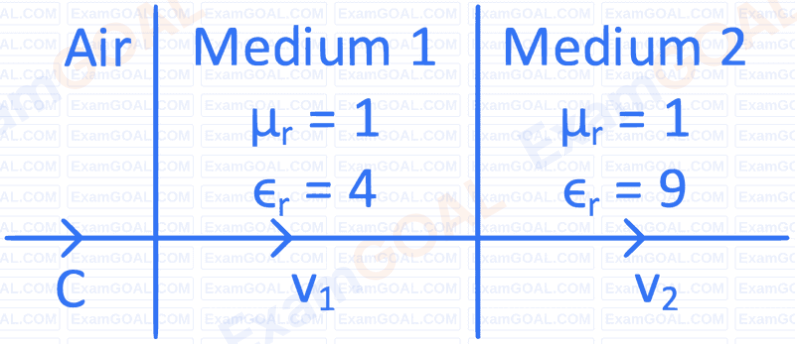
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Q94. JEE Main 2022 (Online) 28th July Morning Shift

Open solution

As shown in the figure, after passing through the medium 1 . The speed of light v_2 in medium 2 will be :

(Given $c = 3 \times 10^8\ ms^{-1}$)



(A) $1.0 \times 10^8\ ms^{-1}$

(B) $0.5 \times 10^8\ ms^{-1}$

(C) $1.5 \times 10^8\ ms^{-1}$

(D) $3.0 \times 10^8\ ms^{-1}$

Q95. JEE Main 2022 (Online) 28th July Morning Shift

Open solution

In normal adjustment, for a refracting telescope, the distance between objective and eye piece is $30\ cm$. The focal length of the objective, when the angular magnification of the telescope is 2 , will be :

(A) 20 cm

(B) 30 cm

(C) 10 cm

(D) 15 cm

Q96. JEE Main 2022 (Online) 27th July Morning Shift

Open solution

A microscope was initially placed in air (refractive index 1). It is then immersed in oil (refractive index 2). For a light whose wavelength in air is λ, calculate the change of microscope's resolving power due to oil and choose the correct option.

(A) Resolving power will be $\frac{1}{4}$ in the oil than it was in the air.

(B) Resolving power will be twice in the oil than it was in the air.

(C) Resolving power will be four times in the oil than it was in the air.

(D) Resolving power will be $\frac{1}{2}$ in the oil than it was in the air.

Q97. JEE Main 2022 (Online) 26th July Evening Shift

Open solution

Light travels in two media M_1 and M_2 with speeds $1.5 \times 10^8\ ms^{-1}$ and $2.0 \times 10^8\ ms^{-1}$ respectively. The critical angle between them is :

(A) $\tan^{-1}\left(\frac{3}{\sqrt{7}}\right)$

(B) $\tan^{-1}\left(\frac{2}{3}\right)$

(C) $\cos^{-1}\left(\frac{3}{4}\right)$

(D) $\sin^{-1}\left(\frac{2}{3}\right)$

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Q98. JEE Main 2022 (Online) 25th July Evening Shift

For an object placed at a distance 2.4 m from a lens, a sharp focused image is observed on a screen placed at a distance 12 cm from the lens. A glass plate of refractive index 1.5 and thickness 1 cm is introduced between lens and screen such that the glass plate plane faces parallel to the screen. By what distance should the object be shifted so that a sharp focused image is observed again on the screen?

Open solution

(A) 0.8 m (B) 3.2 m (C) 1.2 m (D) 5.6 m

Q99. JEE Main 2022 (Online) 25th July Morning Shift

Which of the following statement is correct?

Open solution

(A) In primary rainbow, observer sees red colour on the top and violet on the bottom

(B) In primary rainbow, observer sees violet colour on the top and red on the bottom

(C) In primary rainbow, light wave suffers total internal reflection twice before coming out of water drops.

(D) Primary rainbow is less bright than secondary rainbow.

Q100. JEE Main 2022 (Online) 25th July Morning Shift

Time taken by light to travel in two different materials A and B of refractive indices μ_A and μ_B of same thickness is t_1 and t_2 respectively. If $t_2 - t_1 = 5 \times 10^{-10}$ s and the ratio of μ_A to μ_B is $1 : 2$. Then, the thickness of material, in meter is: (Given v_{A} and v_{B} are velocities of light in A and B materials respectively.)

Open solution

(A) $5 \times 10^{-10}\, v_{\mathrm{A}}\ \mathrm{m}$ (B) $5 \times 10^{-10}\ \mathrm{m}$

(C) $1.5 \times 10^{-10}\ \mathrm{m}$ (D) $5 \times 10^{-10}\, v_{\mathrm{B}}\ \mathrm{m}$

Q101. JEE Main 2022 (Online) 29th June Evening Shift

The speed of light in media 'A' and 'B' are 2.0×10^{10} cm/s and 1.5×10^{10} cm/s respectively. A ray of light enters from the medium B to A at an incident angle 'θ'. If the ray suffers total internal reflection, then

Open solution

(A) $\theta = \sin^{-1}\left(\frac{3}{4}\right)$ (B) $\theta > \sin^{-1}\left(\frac{2}{3}\right)$

(C) $\theta < \sin^{-1}\left(\frac{3}{4}\right)$ (D) $\theta > \sin^{-1}\left(\frac{3}{4}\right)$

Q102. JEE Main 2022 (Online) 28th June Morning Shift

The refracting angle of a prism is A and refractive index of the material of the prism is cot (A/2). Then the angle of minimum deviation will be -

Open solution

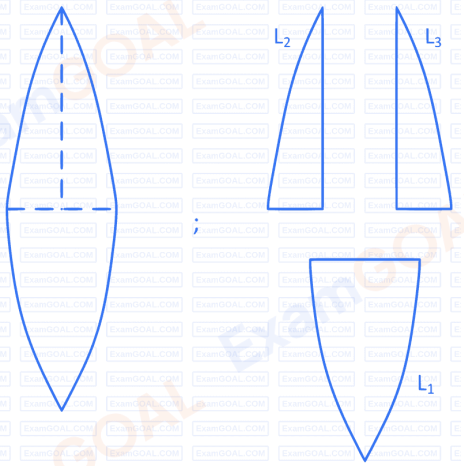
(A) 180 − 2A (B) 90 − A (C) 180 + 2A (D) 180 − 3A

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Q103. JEE Main 2022 (Online) 27th June Evening Shift

A convex lens has power P. It is cut into two halves along its principal axis. Further one piece (out of the two halves) is cut into two halves perpendicular to the principal axis (as shown in figures). Choose the incorrect option for the reported pieces.

Open solution



(A) Power of $L_1 = \frac{P}{2}$

(B) Power of $L_2 = \frac{P}{2}$

(C) Power of $L_3 = \frac{P}{2}$

(D) Power of L_1 = P

Q104. JEE Main 2022 (Online) 27th June Morning Shift

Consider a light ray travelling in air is incident into a medium of refractive index $\sqrt{2n}$. The incident angle is twice that of refracting angle. Then, the angle of incidence will be :

Open solution

(A) $\sin^{-1}\left(\sqrt{n}\right)$

(B) $\cos^{-1}\left(\sqrt{\frac{n}{2}}\right)$

(C) $\sin^{-1}\left(\sqrt{2n}\right)$

(D) $2\cos^{-1}\left(\sqrt{\frac{n}{2}}\right)$

Q105. JEE Main 2022 (Online) 25th June Morning Shift

A light wave travelling linearly in a medium of dielectric constant 4, is incident on the horizontal interface separating medium with air. The angle of incidence for which the total intensity of incident wave will be reflected back into the same medium will be :

(Given : relative permeability of medium μ_r = 1)

Open solution

(A) 10°

(B) 20°

(C) 30°

(D) 60°

Q106. JEE Main 2022 (Online) 25th June Morning Shift

The difference of speed of light in the two media A and B ($v_A - v_B$) is 2.6×10^7 m/s. If the refractive index of medium B is 1.47, then the ratio of refractive index of medium B to medium A is : (Given : speed of light in vacuum c = 3×10^8 ms^{-1})

Open solution

(A) 1.303

(B) 1.318

(C) 1.13

(D) 0.12

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Q107. JEE Main 2021 (Online) 1st September Evening Shift

A glass tumbler having inner depth of 17.5 cm is kept on a table. A student starts pouring water (μ = 4/3) into it while looking at the surface of water from the above. When he feels that the tumbler is half filled, he stops pouring water. Up to what height, the tumbler is actually filled?

Open solution

(A) 11.7 cm (B) 10 cm (C) 7.5 cm (D) 8.75 cm

Q108. JEE Main 2021 (Online) 31st August Morning Shift

An object is placed at the focus of concave lens having focal length f. What is the magnification and distance of the image from the optical centre of the lens?

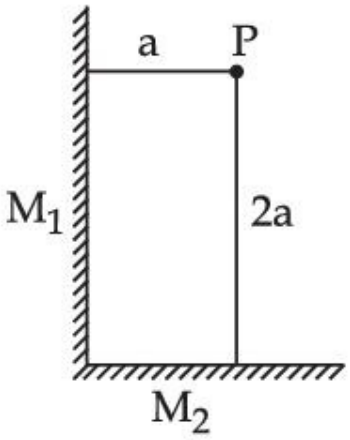
Open solution

(A) 1, ∞ (B) Very high, ∞ (C) $\frac{1}{2}$, $\frac{f}{2}$ (D) $\frac{1}{4}$, $\frac{f}{4}$

Q109. JEE Main 2021 (Online) 31st August Morning Shift

Two plane mirrors M_1 and M_2 are at right angle to each other shown. A point source 'P' is placed at 'a' and '2a' meter away from M_1 and M_2 respectively. The shortest distance between the images thus formed is : (Take $\sqrt{5}$ = 2.3)

Open solution



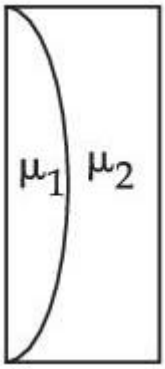
(A) 3a (B) 4.6a (C) 2.3a (D) $2\sqrt{10}$a

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Q110. JEE Main 2021 (Online) 27th August Evening Shift

Open solution

Curved surfaces of a plano-convex lens of refractive index μ_1 and a plano-concave lens of refractive index μ_2 have equal radius of curvature as shown in figure. Find the ratio of radius of curvature to the focal length of the combined lenses.



(A) $\frac{1}{\mu_2-\mu_1}$ (B) $\mu_1-\mu_2$ (C) $\frac{1}{\mu_1-\mu_2}$ (D) $\mu_2-\mu_1$

Q111. JEE Main 2021 (Online) 27th August Morning Shift

Open solution

An object is placed beyond the centre of curvature C of the given concave mirror. If the distance of the object is d_1 from C and the distance of the image formed is d_2 from C, the radius of curvature of this mirror is :

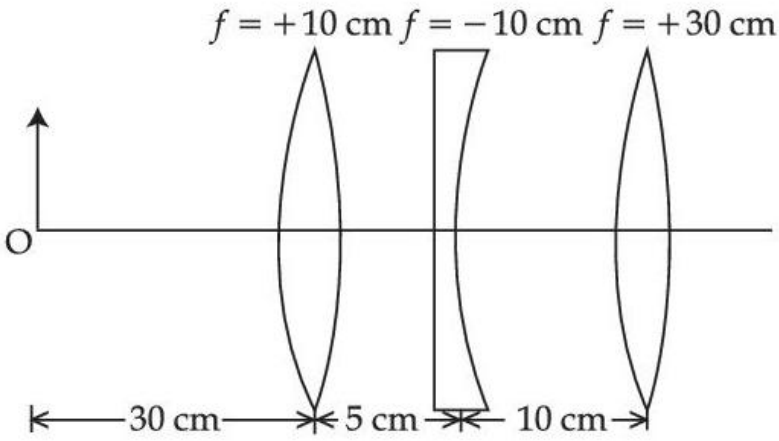
(A) $\frac{2d_1d_2}{d_1-d_2}$ (B) $\frac{2d_1d_2}{d_1+d_2}$

(C) $\frac{d_1d_2}{d_1+d_2}$ (D) $\frac{d_1d_2}{d_1-d_2}$

Q112. JEE Main 2021 (Online) 27th August Morning Shift

Open solution

Find the distance of the image from object O, formed by the combination of lenses in the figure :



(A) 75 cm (B) 10 cm (C) 20 cm (D) infinity

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Q113. JEE Main 2021 (Online) 26th August Morning Shift

Open solution

Car B overtakes another car A at a relative speed of 40 ms^{-1}. How fast will the image of car B appear to move in the mirror of focal length 10 cm fitted in car A, when the car B is 1.9 m away from the car A?

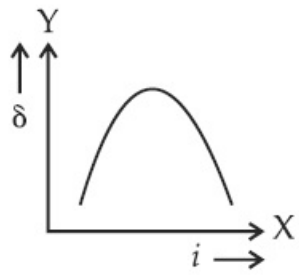
(A) 4 ms^{-1} (B) 0.2 ms^{-1} (C) 40 ms^{-1} (D) 0.1 ms^{-1}

Q114. JEE Main 2021 (Online) 27th July Evening Shift

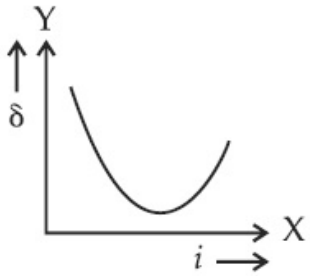
Open solution

The expected graphical representation of the variation of angle of deviation 'δ' with angle of incidence 'i' in a prism is :

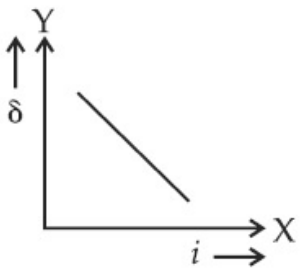
(A)



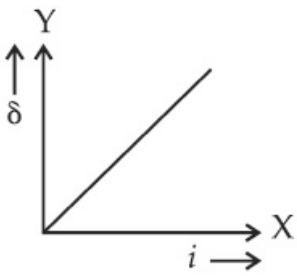
(B)



(C)



(D)



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Q115. JEE Main 2021 (Online) 25th July Evening Shift

Open solution

A prism of refractive index μ and angle of prism A is placed in the position of minimum angle of deviation. If minimum angle of deviation is also A, then the value of A in terms of refractive index is

(A) $2\cos^{-1}\left(\frac{\mu}{2}\right)$

(B) $\sin^{-1}\left(\frac{\mu}{2}\right)$

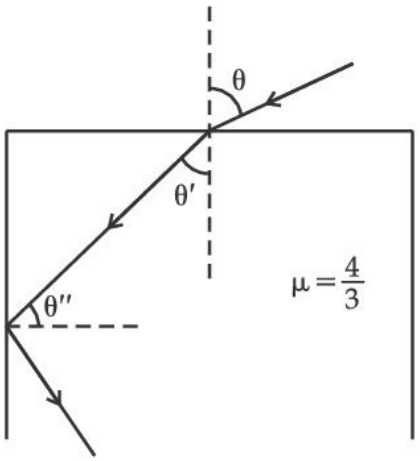
(C) $\sin^{-1}\left(\sqrt{\frac{\mu-1}{2}}\right)$

(D) $\cos^{-1}\left(\frac{\mu}{2}\right)$

Q116. JEE Main 2021 (Online) 25th July Evening Shift

Open solution

A ray of light entering from air into a denser medium of refractive index $\frac{4}{3}$, as shown in figure. The light ray suffers total internal reflection at the adjacent surface as shown. The maximum value of angle θ should be equal to :



(A) $\sin^{-1}\frac{\sqrt{7}}{3}$

(B) $\sin^{-1}\frac{\sqrt{5}}{4}$

(C) $\sin^{-1}\frac{\sqrt{7}}{4}$

(D) $\sin^{-1}\frac{\sqrt{5}}{3}$

Q117. JEE Main 2021 (Online) 25th July Morning Shift

Open solution

A ray of laser of a wavelength 630 nm is incident at an angle of 30° at the diamond-air interface. It is going from diamond to air. The refractive index of diamond is 2.42 and that of air is 1. Choose the correct option.

(A) angle of refraction is 24.41°

(B) angle of refraction is 30°

(C) refraction is not possible

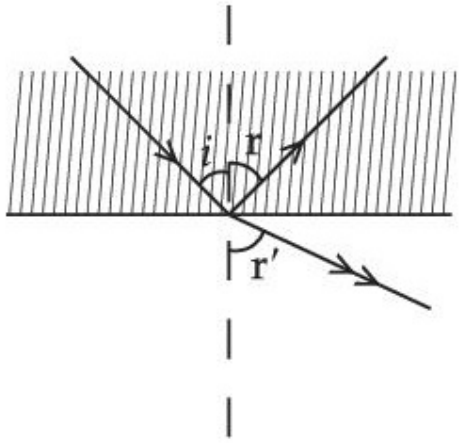
(D) angle of refraction is 53.4°

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Q118. JEE Main 2021 (Online) 22th July Evening Shift

Open solution

A ray of light passes from a denser medium to a rarer medium at an angle of incidence i. The reflected and refracted rays make an angle of 90° with each other. The angle of reflection and refraction are respectively r and r'. The critical angle is given by

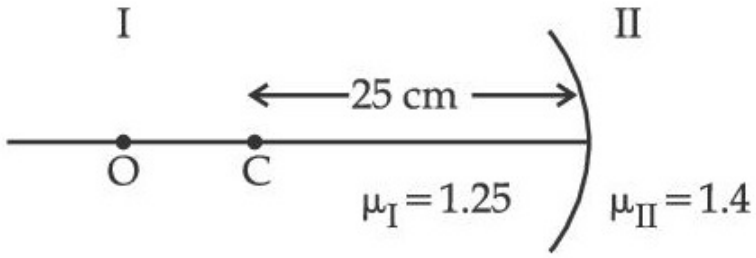


(A) $\sin^{-1}(\tan r)$ (B) $\sin^{-1}(\cot r)$ (C) $\sin^{-1}(\tan r')$ (D) $\tan^{-1}(\sin i)$

Q119. JEE Main 2021 (Online) 20th July Morning Shift

Open solution

Region I and II are separated by a spherical surface of radius 25 cm. An object is kept in region I at a distance of 40 cm from the surface. The distance of the image from the surface is :



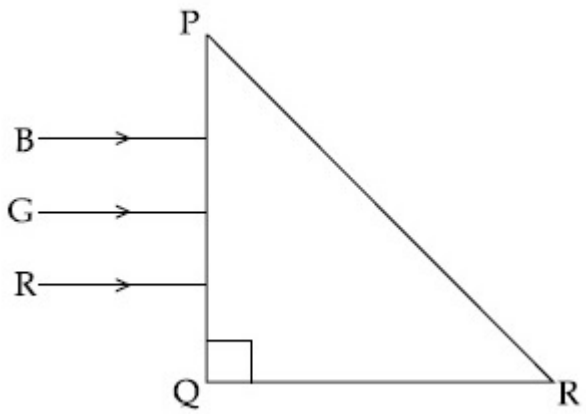
(A) 55.44 cm (B) 18.23 cm (C) 9.52 cm (D) 37.58 cm

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Q120. JEE Main 2021 (Online) 18th March Evening Shift

Three rays of light, namely red (R), green (G) and blue (B) are incident on the face PQ of a right angled prism PQR as shown in the figure.

Open solution



The refractive indices of the material of the prism for red, green and blue wavelength are 1.27, 1.42 and 1.49 respectively. The colour of the ray(s) emerging out of the face PR is :

(A) green (B) blue and green (C) blue (D) red

Q121. JEE Main 2021 (Online) 18th March Morning Shift

Your friend is having eye sight problem. She is not able to see clearly a distant uniform window mesh and it appears to her as non-uniform and distorted. The doctor diagnosed the problem as :

Open solution

(A) Astigmatism (B) Myopia with Astigmatism

(C) Presbyopia with Astigmatism (D) Myopia and Hypermetropia

Q122. JEE Main 2021 (Online) 17th March Morning Shift

The thickness at the centre of a plano-convex lens is 3 mm and the diameter is 6 cm. If the speed of light in the material of the lens is 2×10^8 ms^{-1}, the focal length of the lens is _____.

Open solution

(A) 0.30 cm (B) 30 cm (C) 15 cm (D) 1.5 cm

Q123. JEE Main 2021 (Online) 16th March Evening Shift

The refractive index of a converging lens is 1.4. What will be the focal length of this lens if it is placed in a medium of same refractive index? Assume the radii of curvature of the faces of lens are R_1 and R_2 respectively.

Open solution

(A) Zero (B) Infinite

(C) 1 (D) $\frac{R_1R_2}{R_1-R_2}$

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Q124. JEE Main 2021 (Online) 16th March Evening Shift

Red light differs from blue light as they have :

(A) Different frequencies and same wavelengths

(B) Different frequencies and different wavelengths

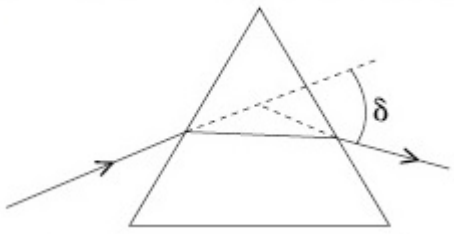
(C) Same frequencies and different wavelengths

(D) Same frequencies and same wavelengths

Open solution

Q125. JEE Main 2021 (Online) 16th March Morning Shift

The angle of deviation through a prism is minimum when



(A) Incident ray and emergent ray are symmetric to the prism.

(B) The refracted ray inside the prism becomes parallel to its base

(C) Angle of incidence is equal to that of the angle of emergence

(D) When angle of emergence is double the angle of incidence

Choose the correct answer from the options given below :

(A) Statements (B) and (C) are true

(B) Only statements (A) and (B) are true

(C) Statements (A), (B) and (C) are true

(D) Only statement (D) is true

Open solution

Q126. JEE Main 2021 (Online) 26th February Evening Shift

The incident ray, reflected ray and the outward drawn normal are denoted by the unit vectors $\vec{a}, \vec{b}$ and $\vec{c}$ respectively. Then choose the correct relation for these vectors.

(A) $\vec{b} = \vec{a} + 2\vec{c}$

(B) $\vec{b} = \vec{a} - 2(\vec{a} \cdot \vec{c})\vec{c}$

(C) $\vec{b} = 2\vec{a} + \vec{c}$

(D) $\vec{b} = \vec{a} - \vec{c}$

Open solution

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Q127. JEE Main 2021 (Online) 26th February Evening Shift

Open solution

Given below are two statements : one is labeled as Assertion A and the other is labeled as Reason R.

Assertion A : For a simple microscope, the angular size of the object equals the angular size of the image.

Reason R : Magnification is achieved as the small object can be kept much closer to the eye than 25 cm and hence it subtends a large angle.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) A is true but R is false

(B) A is false but R is true

(C) Both A and R are true and R is the correct explanation of A

(D) Both A and R are true but R is NOT the correct explanation of A

Q128. JEE Main 2021 (Online) 26th February Morning Shift

Open solution

A short straight object of height 100 cm lies before the central axis of a spherical mirror whose focal length has absolute value | f | = 40 cm. The image of object produced by the mirror is of height 25 cm and has the same orientation of the object. One may conclude from the information :

(A) Image is real, same side of convex mirror.

(B) Image is virtual, opposite side of convex mirror.

(C) Image is virtual, opposite side of concave mirror.

(D) Image is real, same side of concave mirror.

Q129. JEE Main 2021 (Online) 24th February Morning Shift

Open solution

The focal length f is related to the radius of curvature r of the spherical convex mirror by :

(A) f = r

(B) f = $-$ r

(C) f = $+\frac{1}{2}$ r

(D) f = $-\frac{1}{2}$ r

Q130. JEE Main 2020 (Online) 6th September Evening Slot

Open solution

A double convex lens has power P and same radii of curvature R of both the surfaces. The radius of curvature of a surface of a plano-convex lens made of the same material with power 1.5 P is :

(A) $\frac{R}{3}$

(B) $\frac{3R}{2}$

(C) $\frac{R}{2}$

(D) 2R

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Q131. JEE Main 2020 (Online) 6th September Morning Slot

Open solution

A point like object is placed at a distance of 1 m in front of a convex lens of focal length 0.5 m. A plane mirror is placed at a distance of 2 m behind the lens. The position and nature of the final image formed by the system is :

(A) 2.6 m from the mirror, real

(B) 1 m from the mirror, real

(C) 2.6 m from the mirror, virtual

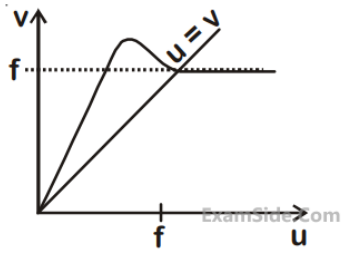
(D) 1 m from the mirror, virtual

Q132. JEE Main 2020 (Online) 5th September Morning Slot

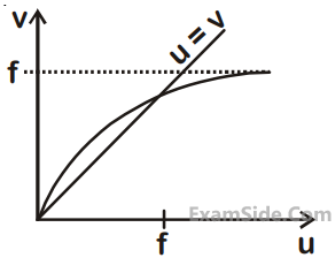
Open solution

For a concave lens of focal length f, the relation between object and image distances u and v, respectively, from its pole can best be represented by (u = v is the reference line) :

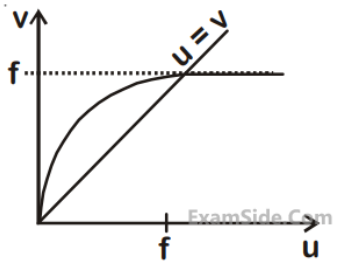
(A)



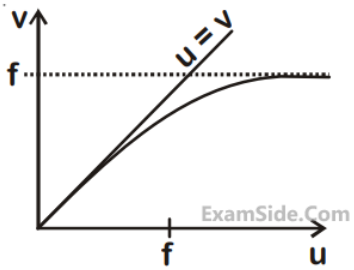
(B)



(C)



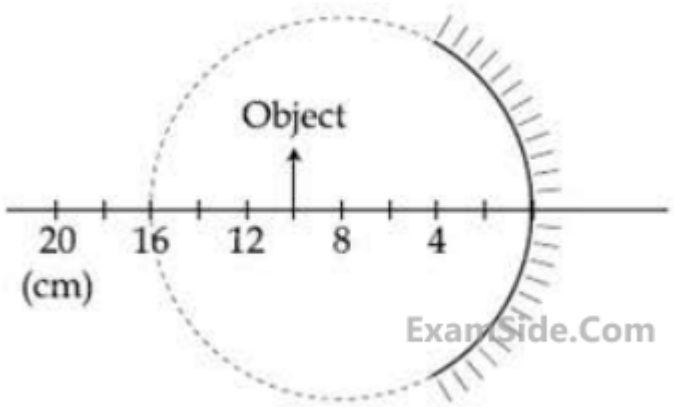
(D)



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Q133. JEE Main 2020 (Online) 2nd September Morning Slot

Open solution



A spherical mirror is obtained as shown in the figure from a hollow glass sphere. If an object is positioned in front of the mirror, what will be the nature and magnification of the image of the object?
(Figure drawn as schematic and not to scale)

(A) Inverted, real and unmagnified
(B) Inverted, real and magnified
(C) Erect, virtual and magnified
(D) Erect, virtual and unmagnified

Q134. JEE Main 2020 (Online) 9th January Evening Slot

Open solution

There is a small source of light at some depth below the surface of water (refractive index = $\frac{4}{3}$) in a tank of large cross sectional surface area. Neglecting any reflection from the bottom and absorption by water, percentage of light that emerges out of surface is (nearly) :
[Use the fact that surface area of a spherical cap of height h and radius of curvature r is $2\pi rh$]:

(A) 17%
(B) 34%
(C) 50%
(D) 21%

Q135. JEE Main 2020 (Online) 9th January Morning Slot

Open solution

A vessel of depth 2h is half filled with a liquid of refractive index $2\sqrt{2}$ and the upper half with another liquid of refractive index $\sqrt{2}$. The liquids are immiscible. The apparent depth of the inner surface of the bottom of vessel will be :

(A) $\frac{h}{\sqrt{2}}$
(B) $\frac{h}{3\sqrt{2}}$
(C) $\frac{3}{4}h\sqrt{2}$
(D) $\frac{h}{2\left(\sqrt{2}+1\right)}$

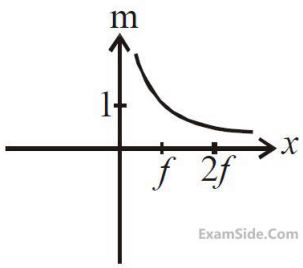
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Q136. JEE Main 2020 (Online) 8th January Evening Slot

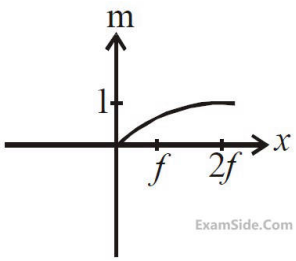
Open solution

An object is gradually moving away from the focal point of a concave mirror along the axis of the mirror. The graphical representation of the magnitude of linear magnification (m) versus distance of the object from the mirror (x) is correctly given by :
(Graphs are drawn schematically and are not to scale)

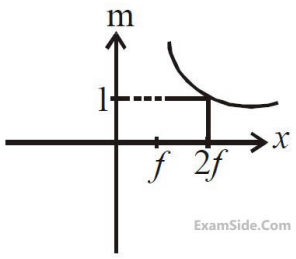
(A)



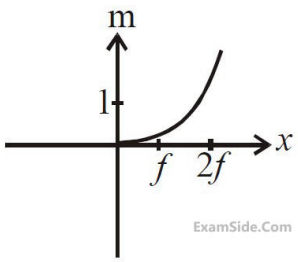
(B)



(C)



(D)



Q137. JEE Main 2020 (Online) 8th January Morning Slot

Open solution

The magnifying power of a telescope with tube 60 cm is 5. What is the focal length of its eye piece ?

(A) 40 cm (B) 10 cm (C) 30 cm (D) 20 cm

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Q138. JEE Main 2020 (Online) 8th January Morning Slot

The critical angle of a medium for a specific wavelength, if the medium has relative permittivity 3 and relative permeability $\frac{4}{3}$ for this wavelength, will be :

(A) 45° (B) 15° (C) 30° (D) 60°

Open solution

Q139. JEE Main 2020 (Online) 7th January Evening Slot

A thin lens made of glass (refractive index = 1.5) of focal length f = 16 cm is immersed in a liquid of refractive index 1.42. If its focal length in liquid is f_1 , then the ratio $\frac{f_1}{f}$ is closest to the integer :

(A) 17 (B) 1 (C) 9 (D) 5

Open solution

Q140. JEE Main 2020 (Online) 7th January Morning Slot

If we need a magnification of 375 from a compound microscope of tube length 150 mm and an objective of focal length 5 mm, the focal length of the eye-piece, should be close to :

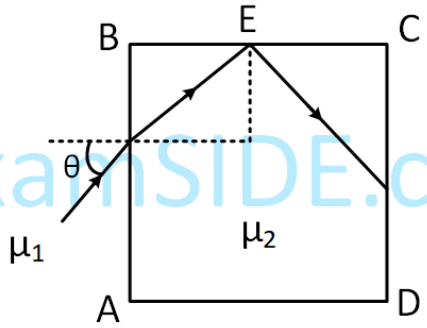
(A) 22 mm (B) 12 mm (C) 33 mm (D) 2 mm

Open solution

Q141. JEE Main 2019 (Online) 12th April Evening Slot

A transparent cube of side, made of a material of refractive index μ_2, is immersed in a liquid of refractive index $\mu_1(\mu_1 < \mu_2)$. A ray is incident on the face AB at an angle θ(shown in the figure). Total internal reflection takes place at point E on the face BC.

Open solution



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Then θ must satisfy :

(A) $\theta > \sin^{-1}\sqrt{\frac{\mu_2^2}{\mu_1^2} - 1}$ (B) $\theta < \sin^{-1}\sqrt{\frac{\mu_2^2}{\mu_1^2} - 1}$

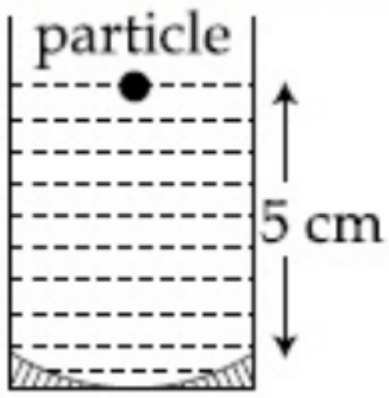
(C) $\theta < \sin^{-1}\frac{\mu_1}{\mu_2}$ (D) $\theta > \sin^{-1}\frac{\mu_1}{\mu_2}$

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Q142. JEE Main 2019 (Online) 12th April Morning Slot

Open solution

A concave mirror has radius of curvature of 40 cm. It is at the bottom of a glass that has water filled up to 5 cm (see figure). If a small particle is floating on the surface of water, its image as seen, from directly above the glass, is at a distance d from the surface of water. The value of d is close to:
(Refractive index of water = 1.33)

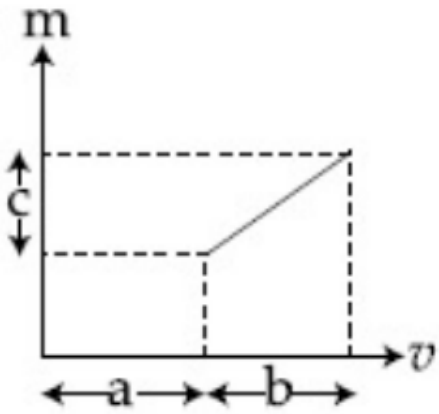


(A) 11.7 cm (B) 6.7 cm (C) 13.4 cm (D) 8.8 cm

Q143. JEE Main 2019 (Online) 10th April Evening Slot

Open solution

The graph shows how the magnification m produced by a thin lens varies with image distance v. What is the focal length of the lens used?



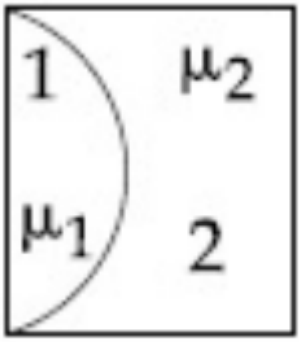
(A) $\frac{b^2}{ac}$ (B) $\frac{b^2c}{a}$ (C) $\frac{a}{c}$ (D) $\frac{b}{c}$

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Q144. JEE Main 2019 (Online) 10th April Morning Slot

Open solution

One plano-convex and one plano-concave lens of same radius of curvature 'R' but of different materials are joined side by side as shown in the figure. If the refractive index of the material of 1 is μ_1 and that of 2 is μ_2, then the focal length of the combination is :



(A) $\frac{2R}{\mu_1-\mu_2}$

(B) $\frac{R}{2-(\mu_1-\mu_2)}$

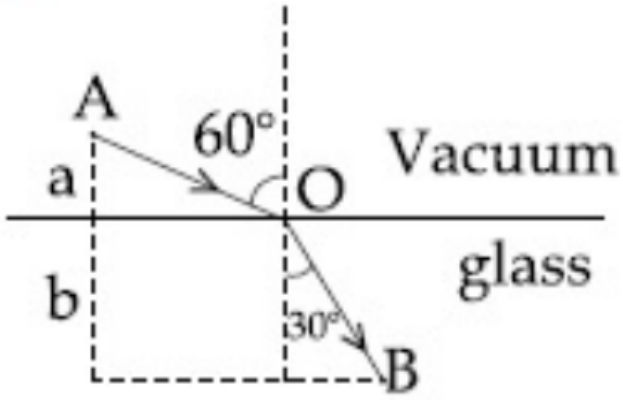
(C) $\frac{R}{\mu_1-\mu_2}$

(D) $\frac{R}{2(\mu_1-\mu_2)}$

Q145. JEE Main 2019 (Online) 10th April Morning Slot

Open solution

A ray of light AO in vacuum is incident on a glass slab at angle 60° and refracted at angle 30° along OB as shown in the figure. The optical path length of light ray from A to B is:



(A) 2a + 2b/$\sqrt{3}$

(B) 2a + 2b/3

(C) 2a + 2b

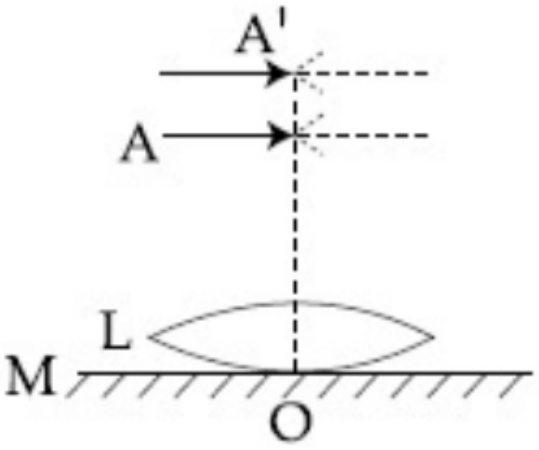
(D) $2\sqrt{3}$/a + 2b

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Q146. JEE Main 2019 (Online) 9th April Evening Slot

Open solution

A thin convex lens L (refractive index = 1.5) is placed on a plane mirror M. When a pin is placed at A, such that OA = 18 cm, its real inverted image is formed at A itself, as shown in figure. When a liquid of refractive index μ1 is put between the lens and the mirror, The pin has to be moved to A', such that OA' = 27 cm, to get its inverted real image at A' itself. The value of μ_1 will be :-



(A) $\sqrt{3}$ (B) 3 / 2 (C) $\sqrt{2}$ (D) 4 / 3

Q147. JEE Main 2019 (Online) 9th April Evening Slot

Open solution

A convex lens of focal length 20 cm produces images of the same magnification 2 when an object is kept at two distances x_1 and x_2 ($x_1 > x_2$) from the lens. The ratio of x_1 and x_2 is :-

(A) 3 : 1 (B) 2 : 1 (C) 5 : 3 (D) 4 : 3

Q148. JEE Main 2019 (Online) 9th April Morning Slot

Open solution

A concave mirror for face viewing has focal length of 0.4 m. The distance at which you hold the mirror from your face in order to see your image upright with a magnification of 5 is :

(A) 0.24 m (B) 0.32 m (C) 1.60 m (D) 0.16 m

Q149. JEE Main 2019 (Online) 8th April Evening Slot

Open solution

A convex lens (of focal length 20 cm) and a concave mirror, having their principal axes along the same lines, are kept 80 cm apart from each other. The concave mirror is to the right of the convex lens. When an object is kept at a distance of 30 cm to the left of the convex lens, its image remains at the same position even if the concave mirror is removed. The maximum distance of the object for which this concave mirror, by itself would produce a virtual image would be :-

(A) 25 cm (B) 10 cm (C) 20 cm (D) 30 cm

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Q150. JEE Main 2019 (Online) 8th April Morning Slot

An upright object is placed at a distance of 40 cm in front of a convergent lens of focal length 20 cm. A convergent mirror of focal length 10 cm is placed at a distance of 60 cm on the other side of the lens. The position and size of the final image will be :

Open solution

(A) 20 cm from the convergent mirror, same size as the object

(B) 40 cm from the convergent mirror, same size as the object

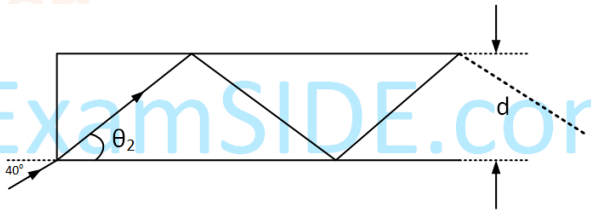
(C) 40 cm from the convergent lens, same size as the object

(D) 20 cm from the convergent mirror, twice the size of the object

Q151. JEE Main 2019 (Online) 8th April Morning Slot

In figure, the optical fiber is l = 2m long and has a diameter of d = 20 μm. If a ray of light is incident on one end of the fiber at angle θ_1 = 40°, the number of reflection it makes before emerging from the other end is close to: (refractive index of fibre is 1.31 and sin 40° = 0.64)

Open solution



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(A) 57000 (B) 55000 (C) 66000 (D) 45000

Q152. JEE Main 2019 (Online) 12th January Evening Slot

A plano-convex lens (focal length f_2, refractive index μ_2, radius of curvature R) fits exactly into a plano-concave lens (focal length f_1, refractive index μ_1, radius of curvature R). Their plane surfaces are parallel to each other. Then, the focal length of the combination will be :

Open solution

(A) $f_1 + f_2$

(B) $f_1 - f_2$

(C) $\frac{R}{\mu_2 - \mu_1}$

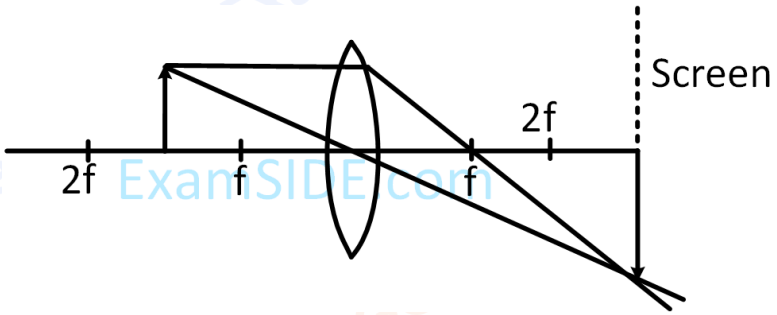
(D) $\frac{2f_1 f_2}{f_1 + f_2}$

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Q153. JEE Main 2019 (Online) 12th January Evening Slot

Open solution

Formation of real image using a biconvex lens is shown below :



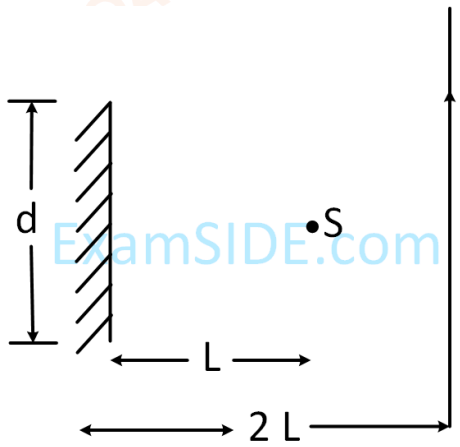
If the whole set up is immersed in water without disturbing the object and the screen positions, what will one observe on the screen ?

(A) Image disappears (B) Magnified image (C) Erect real image (D) No change

Q154. JEE Main 2019 (Online) 12th January Morning Slot

Open solution

A point source of light, S is placed at distance L in front of the centre of plane mirror of width d which is hanging vertically on a wall. A man walks in front of the mirror along a line parallel to the mirror, at a distance 2L as shown below. The distance over which the man can see the image of the light source in the mirror is



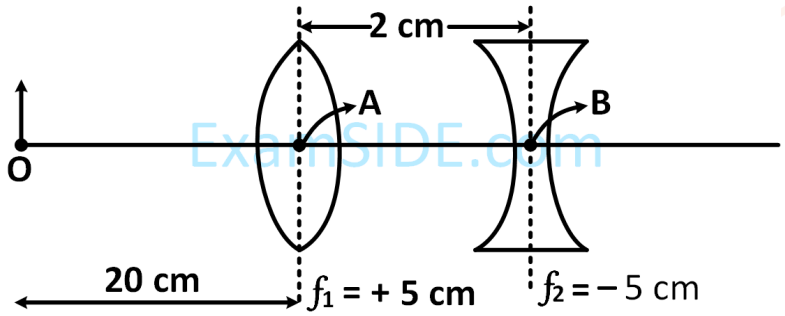
(A) d (B) d/2 (C) 3d (D) 2d

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Q155. JEE Main 2019 (Online) 12th January Morning Slot

What is the position and nature of image formed by lens combination shown in figure ? (f_1, f_2 are focal lengths)

Open solution



(A) $\frac{20}{3}$ cm from point B at right, real

(B) 70 cm from point B at right ; real

(C) 40 cm from point B at right; real

(D) 70 cm from point B at left ; virtual

Q156. JEE Main 2019 (Online) 11th January Evening Slot

A monochromatic light is incident at a certain angle on an equilateral triangular prism and suffers minimum deviation. If the refractive index of the material of the prism is $\sqrt{3}$, then the angle of incidence is:

Open solution

(A) 60°

(B) 45°

(C) 90°

(D) 30°

Q157. JEE Main 2019 (Online) 11th January Morning Slot

An object is at a distance of 20 m from a convex lens of focal length 0.3 m. The lens forms an image of the object. If the object moves away from the lens at a speed of 5 m/s, the speed and direction of the image will be :

Open solution

(A) 1.16×10^{-3} m/s towards the lens

(B) 2.26×10^{-3} m/s away from the lens

(C) 3.22×10^{-3} m/s towards the lens

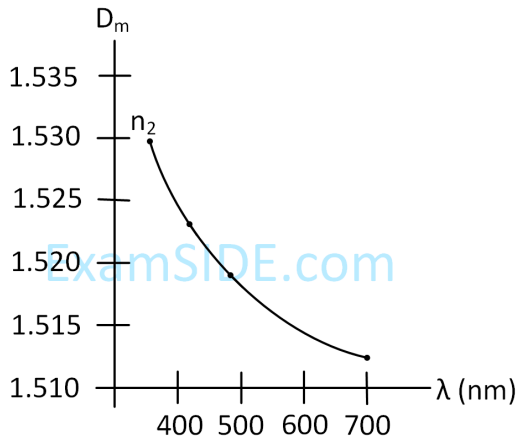
(D) 0.92×10^{-3} m/s away from the lens

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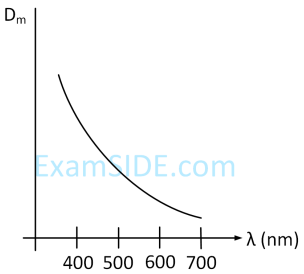
Q158. JEE Main 2019 (Online) 11th January Morning Slot

The variation of refractive index of a crown glass thin prism with wavelength of the incident light is shown. Which of the following graphs is the correct one, if D_m is the angle of minimum deviation?

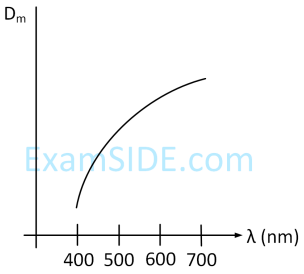
Open solution



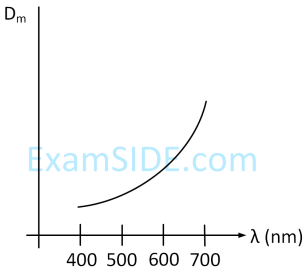
(A)



(B)

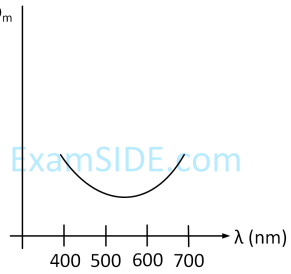


(C)



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(D)



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Q159. JEE Main 2019 (Online) 10th January Evening Slot

The eye can be regarded as a single refracting surface. The radius of curvature of this surface is equal to that of cornea (7.8 mm). This surface separateds two media of refractive indices 1 and 1.34. Calculate the distance from the refracting surface at which a parallel beam of light will come to focus -

Open solution

(A) 2 cm (B) 3.1 cm (C) 4.0 cm (D) 1 cm

Q160. JEE Main 2019 (Online) 10th January Morning Slot

A plano convex lens of refractive index μ_1 and focal length f_1 is kept in contact with another plano concave lens of refractive index μ_2 and focal length f_2. If the radius of curvature of their spherical faces is R each and $f_1 = 2f_2$, then μ_1 and μ_2 are related as -

Open solution

(A) $3\mu_2 - 2\mu_1 = 1$ (B) $\mu_1 + \mu_2 = 3$ (C) $2\mu_1 - \mu_2 = 1$ (D) $2\mu_2 - \mu_1 = 1$

Q161. JEE Main 2019 (Online) 9th January Evening Slot

Two plane mirrors are inclined to each other such that a ray of light incident on the first mirror (M_1) and parallel to the second mirror (M_2) is finally reflected from the second mirror (M_2) parallel to the first mirror (M_1). The angle between the two mirrors will be :

Open solution

(A) 45° (B) 60° (C) 75° (D) 90°

Q162. JEE Main 2019 (Online) 9th January Morning Slot

A convex lens is put 10 cm from a light source and it makes a sharp image on a screen, kept 10 cm from the lens. Now a glass block (refractive index 1.5) of 1.5 cm thickness is placed in contact with the light source. To get the sharp image again, the screen is shifted by a distance d. Then d is :

Open solution

(A) 1.1 cm away from the lens (B) 0
(C) 0.55 cm towards the lens (D) 0.55 cm away from the lens

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Q163. JEE Main 2018 (Online) 16th April Morning Slot

A ray of light is incident at an angle of 60° on one face of a prism of angle 30°. The emergent ray of light makes an angle of 30° with incident ray. The angle made by the emergent ray with second face of prism will be :

Open solution

(A) 0° (B) 90° (C) 45° (D) 30°

Q164. JEE Main 2018 (Online) 15th April Evening Slot

A convergent doublet of separated lenses, corrected for spherical aberration, has resultant focal length of 10 cm. The separation between the two lenses is 2 cm. The focal lengths of the component lenses are :

Open solution

(A) 10 cm, 12 cm (B) 12 cm, 14 cm (C) 16 cm, 18 cm (D) 18 cm, 20cm

Q165. JEE Main 2018 (Online) 15th April Morning Slot

A planoconvex lens becomes an optical system of $28\ cm$ focal length when its plane surface is silvered and illuminated from left to right as shown in Fig-A.

Open solution

If the same lens is instead silvered on the curved surface and illuminated from other side as in Fig-B, it acts like an optical system of focal length $10\ cm$. The refractive index of the material of lens is :



Fig. A Fig. B

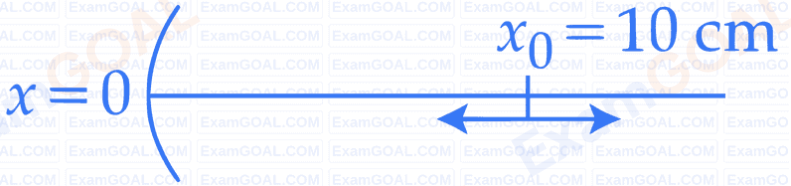
(A) 1.50 (B) 1.55 (C) 1.75 (D) 1.51

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Q166. JEE Main 2018 (Online) 15th April Morning Slot

A particle is oscillating on the X-axis with an amplitude $2\ cm$ about the point $x_0 = 10\ cm$, with a frequency ω. A concave mirror of focal length $5\ cm$ is placed at the origin (see figure).

Open solution



Identify the **correct** statements.

(A) The image executes periodic motion.

(B) The image executes non-periodic motion.

(C) The turning points of the image are asymmetric w.r.t. the image of the point at $x = 10\ cm$.

(D) The distance between the turning points of the oscillation of the image is $\frac{100}{21}\ cm$.

(A) (A), (D) **(B)** (A), (C), (D) **(C)** (B), (D) **(D)** (B), (C)

Q167. JEE Main 2017 (Online) 9th April Morning Slot

In an experiment a convex lens of focal length 15 cm is placed coaxially on an optical bench in front of a convex mirror at a distance of 5 cm from it. It is found that an object and its image coincide, if the object is placed at a distance of 20 cm from the lens. The focal length of the convex mirror is :

Open solution

(A) 27.5 cm **(B)** 20.0 cm **(C)** 25.0 cm **(D)** 30.5 cm

Q168. JEE Main 2017 (Online) 8th April Morning Slot

Let the refractive index of a denser medium with respect to a rarer medium be n_{12} and its critical angle be θ_C . At an angle of incidence A when light is travelling from denser medium to rarer medium, a part of the light is reflected and the rest is refracted and the angle between reflected and refracted rays is 90°. Angle A is given by :

Open solution

(A) $\frac{1}{\cos^{-1}(\sin\theta_C)}$ **(B)** $\frac{1}{\tan^{-1}(\sin\theta_C)}$

(C) $\cos^{-1}(\sin\theta_C)$ **(D)** $\tan^{-1}(\sin\theta_C)$

Q169. JEE Main 2017 (Offline)

A diverging lens with magnitude of focal length 25 cm is placed at a distance of 15cm from a converging lens of magnitude of focal length 20cm. A beam of parallel light falls on the diverging lens. The final image formed is:

Open solution

(A) real and at a distance of 6 cm from the convergent lens.

(B) real and at a distance of 40 cm from convergent lens.

(C) virtual and at a distance of 40 cm from convergent lens.

(D) real and at a distance of 40 cm from the divergent lens.

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Q170. JEE Main 2016 (Online) 10th April Morning Slot

To determine refractive index of glass slab using a travelling microscope, minimum number of readings required are :

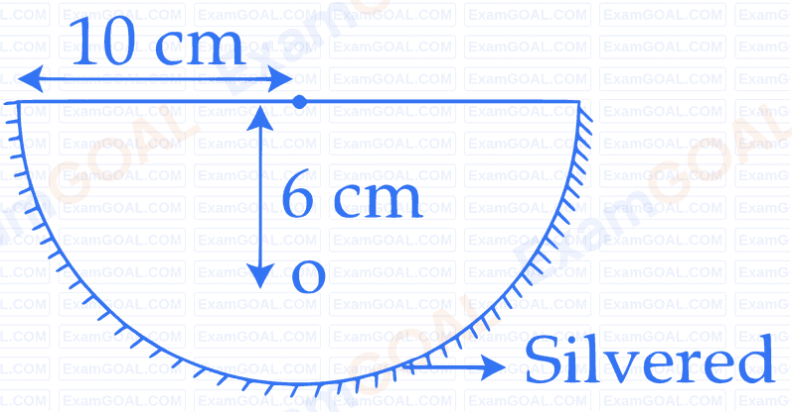
(A) Two (B) Three (C) Four (D) Five

Open solution

Q171. JEE Main 2016 (Online) 10th April Morning Slot

A hemispherical glass body of radius 10 cm and refractive index 1.5 is silvered on its curved surface. A small air bubble is 6 cm below the flat surface inside it along the axis. The position of the image of the air bubble made by the mirror is seen :

Open solution



(A) 14 cm below flat surface (B) 30 cm below flat surface

(C) 20 cm below flat surface (D) 16 cm below flat surface

Q172. JEE Main 2016 (Online) 9th April Morning Slot

To find the focal length of a convex mirror, a student records the following data :

Open solution

Object Pin	Convex Lens	Convex Mirror	Image Pin
22.2 cm	32.2 cm	45.8 cm	71.2 cm

The focal length of the convex lens is f_1 and that of mirror is f_2. Then taking index correction to be negligibly small, f_1 and f_2 are close to :

(A) f_1 = 12.7 cm f_2 = 7.8 cm (B) f_1 = 7.8 cm f_2 = 12.7 cm

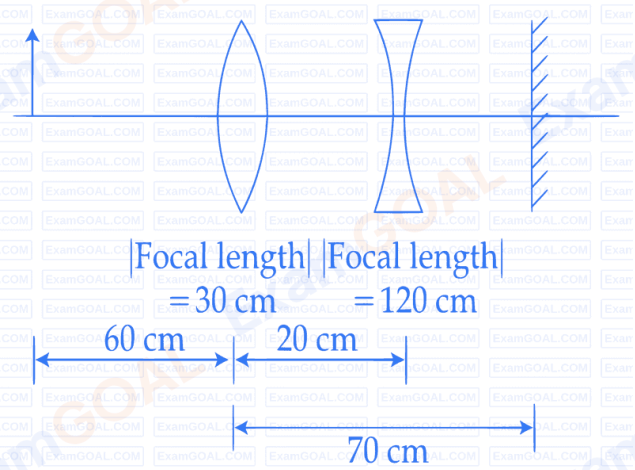
(C) f_1 = 7.8 cm f_2 = 25.4 cm (D) f_1 = 15.6 cm f_2 = 25.4 cm

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Q173. JEE Main 2016 (Online) 9th April Morning Slot

A convex lens, of focal length 30 cm, a concave lens of focal length 120 cm, and aplane mirror are arranged as shown. For an object kept at a distance of 60 cm from the convex lens, the final image, formed by the combination, is a real image, at a distance of :

Open solution



(A) 60 cm from the convex lens
(B) 60 cm from the concave lens
(C) 70 cm from the convex lens
(D) 70 cm from the concave lens

Q174. JEE Main 2016 (Offline)

An observer looks at a distant tree of height $10\ m$ with a telescope of magnifying power of 20. To the observer the tree appears:

(A) 20 times taller
(B) 20 times nearer
(C) 10 times taller
(D) 10 times nearer

Open solution

Q175. JEE Main 2016 (Offline)

In an experiment for determination of refractive index of glass of a prism by $i - \delta$ plot, it was found that a ray incident at angle 35°, suffers a deviation of 40° and that it emerges at angle 79°. In that case which of the following is closest to the maximum possible value of the refractive index?

(A) 1.7
(B) 1.8
(C) 1.5
(D) 1.6

Open solution

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Q176. JEE Main 2015 (Online) 11th April Morning Slot

A thin convex lens of focal length 'f' is put on a plane mirror as shown in the figure. When an object is kept at a distance 'a' from the lens – mirror combination, its image is formed at a distance $\frac{a}{3}$ in front of the combination. The value of 'a' is :

Open solution



(A) f (B) 2f (C) 3f (D) $\frac{3}{2}f$

Q177. JEE Main 2015 (Online) 10th April Morning Slot

You are asked to design a shaving mirror assuming that a person keeps it 10 cm from his face and views the magnified image of the face at the closest comfortable distance of 25 cm. The radius of curvature of the mirror would then be :

Open solution

(A) 30 cm (B) 24 cm (C) 60 cm (D) – 24 cm

Q178. JEE Main 2015 (Online) 10th April Morning Slot

A telescope has an objective lens of focal length 150 cm and an eyepiece of focal length 5 cm. If a 50 m tall tower at a distance of 1 km is observed through this telescope in normal setting, the angle formed by the image of the tower is θ, then θ is close to :

Open solution

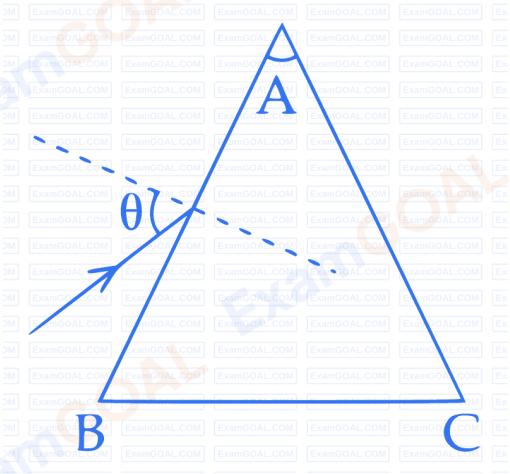
(A) 1° (B) 15° (C) 30° (D) 60°

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Q179. JEE Main 2015 (Offline)

Open solution

Monochromatic light is incident on a glass prism of angle A. If the refractive index of the material of the prism is μ, a ray, incident at an angle θ. on the face AB would get transmitted through the face AC of the prism provided :



(A) $\theta > \cos^{-1}\left[\mu \sin\left(A + \sin^{-1}\left(\frac{1}{\mu}\right)\right)\right]$

(B) $\theta < \cos^{-1}\left[\mu \sin\left(A + \sin^{-1}\left(\frac{1}{\mu}\right)\right)\right]$

(C) $\theta > sin^{-1}\left[\mu \sin\left(A - \sin^{-1}\left(\frac{1}{\mu}\right)\right)\right]$

(D) $\theta < sin^{-1}\left[\mu \sin\left(A - \sin^{-1}\left(\frac{1}{\mu}\right)\right)\right]$

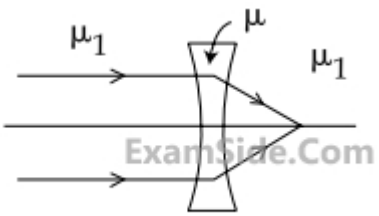
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Q180. JEE Main 2014 (Online) 12th April Morning Slot

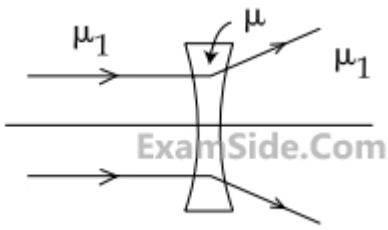
Open solution

The refractive index of the material of a concave lens is μ. It is immersed in a medium of refractive index μ_1. A parallel beam of light is incident on the lens. The path of the emergent rays when $\mu_1 > \mu$ is :

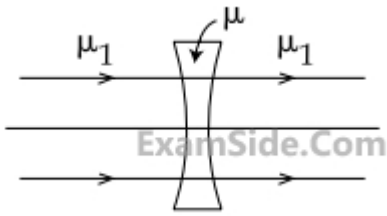
(A)



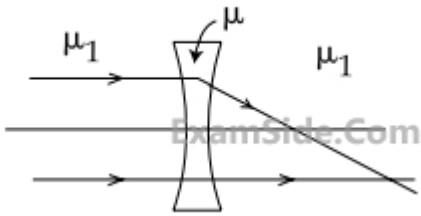
(B)



(C)



(D)



Q181. JEE Main 2014 (Online) 11th April Morning Slot

Open solution

An object is located in a fixed position in front of a screen, Sharp image is obtained on the screen for two positions of a thin lens separated by 10 cm. The size of the images in two situations are in the ratio 3 : 2. What is the distance between the screen and the object ?

(A) 124.5 cm (B) 144.5 cm (C) 65.0 cm (D) 99.0 cm

Q182. JEE Main 2014 (Online) 11th April Morning Slot

Open solution

In a compound microscope the focal length of objective lens is 1.2 cm and focal length of eye piece is 3.0 cm. When object is kept at 1.25 cm in front of objective, final image is formed at infinity. Magnifying power of the compound microscope should be :

(A) 200 (B) 100 (C) 400 (D) 150

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Q183. JEE Main 2014 (Online) 9th April Morning Slot

A diver looking up through the water sees the outside world contained in a circular horizon. The refractive index of water is $\frac{4}{3}$ and the diver's eyes are 15 cm below the surface of water. Then the radius of the circle is :

Open solution

(A) $15 \times 3 \times \sqrt{5}$ cm

(B) $15 \times 3\sqrt{7}$ cm

(C) $\frac{15 \times \sqrt{7}}{3}$ cm

(D) $\frac{15 \times 3}{\sqrt{7}}$ cm

Q184. JEE Main 2014 (Online) 9th April Morning Slot

The focal lengths of objective lens and eye lens of a Galilean Telescope are respectively 30 cm and 3.0 cm. Telescope produces virtual, erect image of an object situated far away from it at least distance of distinct vision from the eye lens. In this condition, the Magnifying Power of the Galilean Telescope should be :

Open solution

(A) + 11.2 (B) − 11.2 (C) − 8.8 (D) + 8.8

Q185. JEE Main 2014 (Offline)

A thin convex lens made from crown glass $\left(\mu = \frac{3}{2}\right)$ has focal length f. When it is measured in two different liquids having refractive indices $\frac{4}{3}$ and $\frac{5}{3}$, it has the focal lengths f_1 and f_2 respectively. The correct relation between the focal lengths is :

Open solution

(A) $f_1 = f_2 < f$

(B) $f_1 > f$ and f_2 becomes negative

(C) $f_2 > f$ and f_1 becomes negative

(D) f_1 and f_2 both become negative

Q186. JEE Main 2014 (Offline)

A green light is incident from the water to the air - water interface at the critical angle (θ). Select the correct statement.

Open solution

(A) The entire spectrum of visible light will come out of the water at an angle of 90° to the normal.

(B) The spectrum of visible light whose frequency is less than that of green light will come out to the air medium.

(C) The spectrum of visible light whose frequency is more than that of green light will come out to the air medium.

(D) The entire spectrum of visible light will come out of the water at various angles to the normal.

Q187. JEE Main 2013 (Online) 25th April Morning Slot

A printed page is pressed by a glass of water. The refractive index of the glass and water is 1.5 and 1.33, respectively. If the thickness of the bottom of glass is 1 cm and depth of water is 5 cm, how much the page will appear to be shifted if viewed from the top?

Open solution

(A) 1.033 cm (B) 3.581 cm (C) 1.3533 cm (D) 1.90 cm

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Q188. JEE Main 2013 (Online) 23th April Morning Slot

Open solution

This question has Statement-1 and Statement-2. Of the four choices given after the Statements, choose the one that best describes the two Statements.
Statement 1: Very large size telescopes are reflecting telescopes instead of refracting telescopes.
Statement 2: It is easier to provide mechanical support to large size mirrors than large size lenses.

(A) Statement-1 is true and Statement-2 is false.

(B) Statement-1 is false and Statement-2 is true.

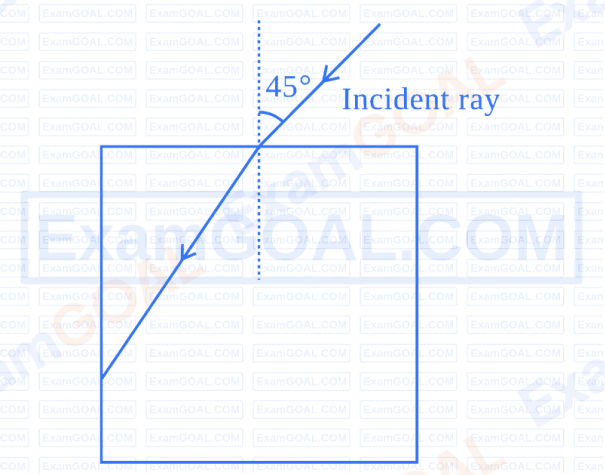
(C) Statements-1 and statement-2 are true and Statement-2 is correct explanation for statement-1.

(D) Statements-1 and statement-2 are true and Statement-2 is not the correct explanation for statement-1.

Q189. JEE Main 2013 (Online) 23th April Morning Slot

Open solution

A light ray falls on a square glass slab as shown in the diagram. The index of refraction of the glass, if total internal reflection is to occur at the vertical face, is equal to :



(A) $\frac{(\sqrt{2}+1)}{2}$ (B) $\sqrt{\frac{5}{2}}$ (C) $\frac{3}{2}$ (D) $\sqrt{\frac{3}{2}}$

Q190. JEE Main 2013 (Online) 22th April Morning Slot

Open solution

The focal length of the objective and the eyepiece of a telescope are 50 cm and 5 cm respectively. If the telescope is focussed for distinct vision on a scale distant 2 m from its objective, then its magnifying power will be :

(A) -4 (B) -8 (C) +8 (D) -2

Q191. JEE Main 2013 (Online) 22th April Morning Slot

Open solution

The image of an illuminated square is obtained on a screen with the help of a converging lens. The distance of the square from the lens is 40 cm. The area of the image is 9 times that of the square. The focal length of the lens is :

(A) 36 cm (B) 27 cm (C) 60 cm (D) 30 cm

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Q192. JEE Main 2013 (Online) 9th April Morning Slot

Light is incident from a medium into air at two possible angles of incidence (A) 20° and (B) 40°. In the medium light travels 3.0 cm in 0.2 ns. The ray will :

Open solution

(A) suffer total internal reflection in both cases (A) and (B)

(B) suffer total internal reflection in case (B) only

(C) have partial reflection and partial transmission in case (B)

(D) have 100% transmission in case (A)

Q193. JEE Main 2013 (Offline)

Diameter of a plano-convex lens is $6\ cm$ and thickness at the center is $3mm$. If speed of light in material of lens is $2 \times 10^8\ m/s$, the focal length of the lens is

Open solution

(A) $15\ cm$ (B) $20\ cm$ (C) $30\ cm$ (D) $10\ cm$

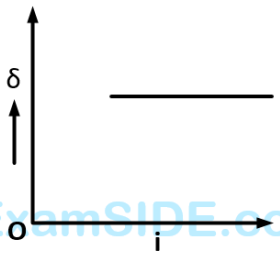
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Q194. JEE Main 2013 (Offline)

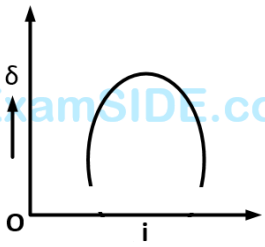
Open solution

The graph between angle of deviation (δ) and angle of incidence (i) for a triangular prism is represented by

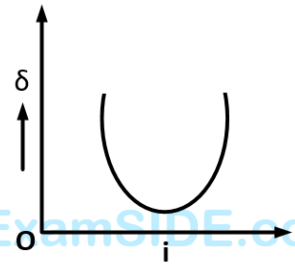
(A)



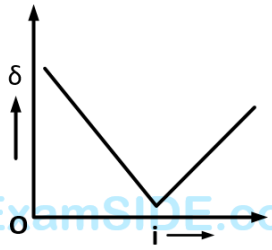
(B)



(C)



(D)

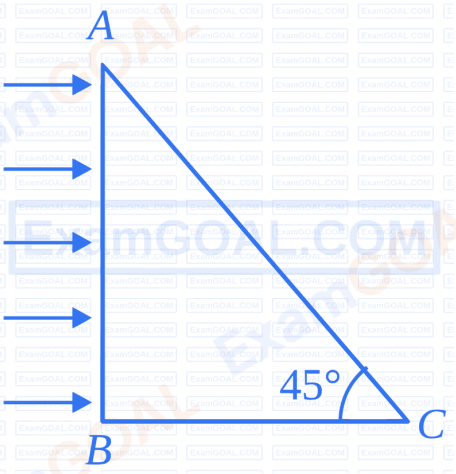


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Q195. AIEEE 2012 (Online) 26th May

Open solution

A beam of light consisting of red, green and blue colours is incident on a right-angled prism on face AB. The refractive indices of the material for the above red, green and blue colours are 1.39, 1.44 and 1.47 respectively.



A person looking on surface AC of the prism will see

(A) no light

(B) green and blue colours

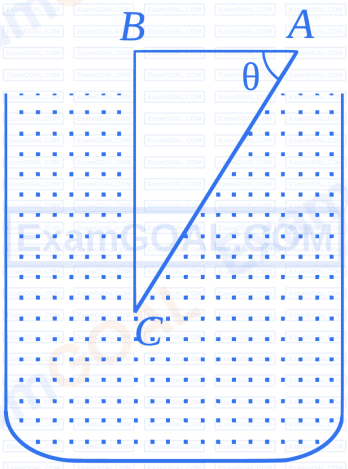
(C) red and green colours

(D) red colour only

Q196. AIEEE 2012 (Online) 19th May

Open solution

A glass prism of refractive index 1.5 is immersed in water (refractive index $\frac{4}{3}$) as shown in figure. A light beam incident normally on the face AB is totally reflected to reach the face BC, if



(A) $\sin\theta > \frac{5}{9}$

(B) $\sin\theta > \frac{2}{3}$

(C) $\sin\theta > \frac{8}{9}$

(D) $\sin\theta > \frac{1}{3}$

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Q197. AIEEE 2012 (Online) 12th May

We wish to make a microscope with the help of two positive lenses both with a focal length of $20\ \text{mm}$ each and the object is positioned $25\ \text{mm}$ from the objective lens. How far apart the lenses should be so that the final image is formed at infinity?

Open solution

(A) $20\ \text{mm}$ (B) $100\ \text{mm}$ (C) $120\ \text{mm}$ (D) $80\ \text{mm}$

Q198. AIEEE 2012 (Online) 7th May

Which of the following processes play a part in the formation of a rainbow?
(i) Refraction
(ii) Total internal reflection
(iii) Dispersion
(iv) Interference

Open solution

(A) (i), (ii) and (iii) (B) (i) and (ii)
(C) (i), (ii) and (iv) (D) (iii) and (iv)

Q199. AIEEE 2012

An object $2.4\,m$ in front of a lens forms a sharp image on a film $12\ cm$ behind the lens. A glass plate $1\ cm$ thick, of refractive index 1.50 is interposed between lens and film with its plane faces parallel to film. At what distance (from lens) should object shifted to be in sharp focus of film?

Open solution

(A) $7.2\ m$ (B) $24\ m$ (C) $3.2\ m$ (D) $5.6\ m$

Q200. AIEEE 2011

Let x-y plane be the boundary between two transparent media. Medium 1 in $z \geq 0$ has a refractive index of $\sqrt{2}$ and medium 2 with $z < 0$ has a refractive index of $\sqrt{3}$. A ray of light in medium 1 given by the vector $\vec{A} = 6\sqrt{3}\hat{i} + 8\sqrt{3}\hat{j} - 10\hat{k}$ is incident on the plane of separation. The angle of refraction in medium 2 is:

Open solution

(A) 45° (B) 60° (C) 75° (D) 30°

Q201. AIEEE 2011

A car is fitted with a convex side-view mirror of focal length $20\ cm$. A second car $2.8m$ behind the first car is overtaking the first car at a relative speed of $15\ m/s$. The speed of the image of the second car as seen in the mirror of the first one is :

Open solution

(A) $\frac{1}{15}\ m/s$ (B) $10\,m/s$ (C) $15\,m/s$ (D) $\frac{1}{10}\ m/s$

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Q202. AIEEE 2010

An initially parallel cylindrical beam travels in a medium of refractive index $\mu(I) = \mu_0 + \mu_2 I$, where μ_0 and μ_2 are positive constants and I is the intensity of the light beam. The intensity of the beam is decreasing with increasing radius.
As the beam enters the medium, it will

Open solution

(A) diverge

(B) converge

(C) diverge near the axis and converge near the periphery

(D) travel as a cylindrical beam

Q203. AIEEE 2010

An initially parallel cylindrical beam travels in a medium of refractive index $\mu(I) = \mu_0 + \mu_2 I$, where μ_0 and μ_2 are positive constants and I is the intensity of the light beam. The intensity of the beam is decreasing with increasing radius.
The speed of light in the medium is

Open solution

(A) minimum on the axis of the beam

(B) the same everywhere in the beam

(C) directly proportional to the intensity I

(D) maximum on the axis of the beam

Q204. AIEEE 2009

In an optics experiment, with the position of the object fixed, a student varies the position of a convex lens and for each position, the screen is adjusted to get a clear image of the object. A graph between the object distance u and the image distance v, from the lens, is plotted using the same scale for the two axes. A straight line passing through the origin and making an angle of 45° with the x-axis meets the experimental curve at P. The coordinates of P will be :

Open solution

(A) $\left(\frac{f}{2}, \frac{f}{2}\right)$

(B) (f, f)

(C) $(4f, 4f)$

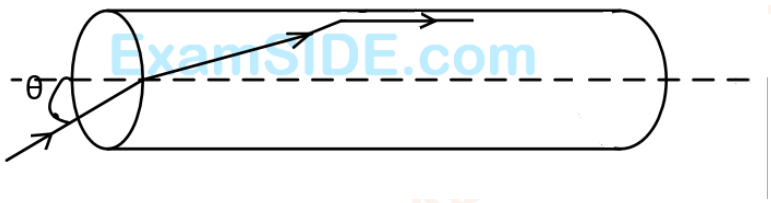
(D) $(2f, 2f)$

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Q205. AIEEE 2009

A transparent solid cylindrical rod has a refractive index of $\frac{2}{\sqrt{3}}$. It is surrounded by air. A light ray is incident at the mid-point of one end of the rod as shown in the figure.

Open solution



The incident angle θ for which the light ray grazes along the wall of the rod is :

(A) $\sin^{-1}\left(\sqrt{3}/2\right)$

(B) $\sin^{-1}\left(2/\sqrt{3}\right)$

(C) $\sin^{-1}\left(\frac{1}{\sqrt{3}}\right)$

(D) $\sin^{-1}\left(1/2\right)$

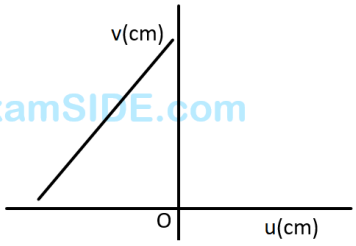
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Q206. AIEEE 2008

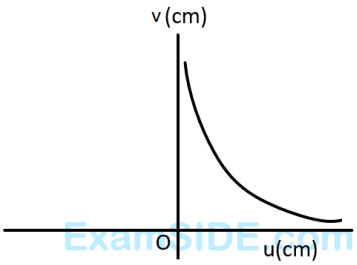
A student measures the focal length of a convex lens by putting an object pin at a distance $'u'$ from the lens and measuring the distance $'v'$ of the image pin. The graph between $'u'$ and $'v'$ plotted by the student should look like

Open solution

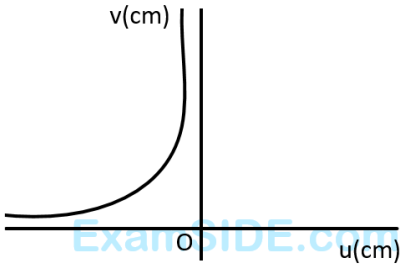
(A)



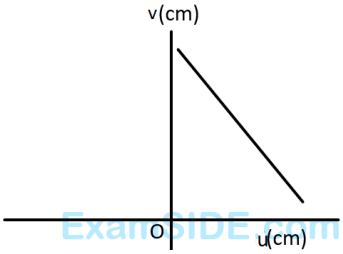
(B)



(C)



(D)



Q207. AIEEE 2007

Two lenses of power $-15\ D$ and $+5\ D$ are in contact with each other. The focal length of the combination is

Open solution

(A) $+10\ cm$ (B) $-20\ cm$ (C) $-10\ cm$ (D) $+20\ cm$

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Q208. AIEEE 2006

The refractive index of a glass is 1.520 for red light and 1.525 for blue light. Let D_1 and D_2 be angles of minimum deviation for red and blue light respectively in a prism of this glass. Then,

Open solution

(A) $D_1 < D_2$

(B) $D_1 = D_2$

(C) D_1 can be less than or greater than D_2 depending upon the angle of prism

(D) $D_1 > D_2$

Q209. AIEEE 2005

A fish looking up through the water sees the outside world contained in a circular horizon. If the refractive index of water is $\frac{4}{3}$ and the fish is $12\ cm$ below the surface, the radius of this circle in cm is

Open solution

(A) $\frac{36}{\sqrt{7}}$ (B) $36\sqrt{7}$ (C) $4\sqrt{5}$ (D) $36\sqrt{5}$

Q210. AIEEE 2005

A thin glass (refractive index 1.5) lens has optical power of $-5\ D$ in air. Its optical power in a liquid medium with refractive index 1.6 will be

Open solution

(A) $-1\ D$ (B) $1\ D$ (C) $-25\ D$ (D) $25\ D$

Q211. AIEEE 2004

A plano convex lens of refractive index 1.5 and radius of curvature $30\ cm$ is silvered at the curved surface. Now this lens has been used to form the image of an object. At what distance from this lens an object be placed in order to have a real image of size of the object

Open solution

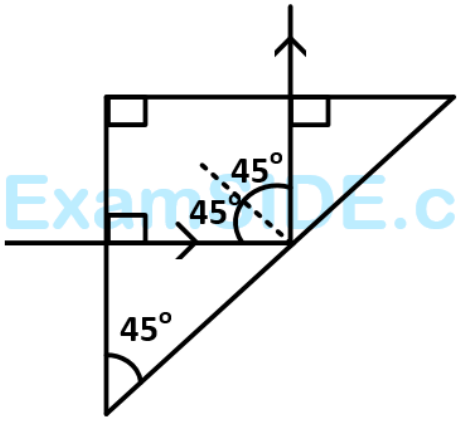
(A) $60\ cm$ (B) $30\ cm$ (C) $20\ cm$ (D) $80\ cm$

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Q212. AIEEE 2004

A light ray is incident perpendicularly to one face of a 90° prism and is totally internally reflected at the glass-air interface. If the angle of reflection is 45°, we conclude that the refractive index n

Open solution



(A) $n > \frac{1}{\sqrt{2}}$ (B) $n > \sqrt{2}$ (C) $n < \frac{1}{\sqrt{2}}$ (D) $n < \sqrt{2}$

Q213. AIEEE 2003

The image formed by an objective of a compound microscope is

(A) virtual and diminished (B) real an diminished

(C) real and enlarged (D) virtual and enlarged

Open solution

Q214. AIEEE 2003

To get three images of a single object, one should have two plane mirrors at an angle of

(A) 60° (B) 90° (C) 120° (D) 30°

Open solution

Q215. AIEEE 2002

An astronomical telescope has a large aperture to

(A) reduce spherical aberration (B) have high resolution

(C) increase span of observation (D) have low dispersion

Open solution

Q216. AIEEE 2002

If two mirrors are kept at 60° to each other, then the number of images formed by them is

(A) 5 (B) 6 (C) 7 (D) 8

Open solution

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Q217. AIEEE 2002

Which of the following is used in optical fibres?

(A) total internal reflection
(B) scattering
(C) diffracttion
(D) refraction

Open solution

Numerical 68 questions

Q218. JEE Main 2026 (Online) 6th April Evening Shift

A concave mirror of focal length 10 cm forms an image which is double the size of object when the object is placed at two different positions. The distance between the two positions of the object is ____ cm.

Open solution

Q219. JEE Main 2026 (Online) 2nd April Evening Shift

If sunlight is focused on a paper using convex lens, it starts burning the paper in shortest time when the lens is kept at 30 cm above the paper. If the radius of curvature of the lens is 60 cm then the refractive index of the lens material is $\frac{\alpha}{10}$. The value of α is _____.

Open solution

Q220. JEE Main 2026 (Online) 28th January Morning Shift

A convex lens of refractive index 1.5 and focal length $f = 18\ \mathrm{cm}$ is immersed in water. The difference in focal lengths of the given lens when it is in water and in air is $\alpha \times \mathrm{f}$. The value of α is ____ .

(refractive index of water $= 4/3$)

Open solution

Q221. JEE Main 2026 (Online) 23rd January Evening Shift

The size of the images of an object, formed by a thin lens are equal when the object is placed at two different positions 8 cm and 24 cm from the lens. The focal length of the lens is ____ cm.

Open solution

Q222. JEE Main 2026 (Online) 22nd January Morning Shift

A parallel beam of light travelling in air (refractive index 1.0) is incident on a convex spherical glass surface of radius of curvature 50 cm . Refractive index of glass is 1.5 . The rays converge to a point at a distance $x\ \mathrm{cm}$ from the centre of the curvature of the spherical surface. The value of x is ____ cm .

Open solution

Q223. JEE Main 2026 (Online) 21st January Morning Shift

In a microscope the objective is having focal length $f_o = 2\ \mathrm{cm}$ and eye-piece is having focal length $f_e = 4\ \mathrm{cm}$. The tube length is 32 cm . The magnification produced by this microscope for normal adjustment is ____ .

Open solution

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Q224. JEE Main 2026 (Online) 21st January Morning Shift

A collimated beam of light of diameter 2 mm is propagating along x-axis. The beam is required to be expanded in a collimated beam of diameter 14 mm using a system of two convex lenses. If first lens has focal length 40 mm , then the focal length of second lens is ____ mm.

Open solution

Q225. JEE Main 2025 (Online) 7th April Morning Shift

A container contains a liquid with refractive index of 1.2 up to a height of 60 cm and another liquid having refractive index 1.6 is added to height H above first liquid. If viewed from above, the apparent shift in the position of bottom of container is 40 cm . The value of H is _____ cm . (Consider liquids are immisible)

Open solution

Q226. JEE Main 2025 (Online) 4th April Morning Shift

Distance between object and its image (magnified by $-\frac{1}{3}$) is 30 cm . The focal length of the mirror used is $\left(\frac{x}{4}\right)\mathrm{cm}$, where magnitude of value of x is _____.

Open solution

Q227. JEE Main 2025 (Online) 3rd April Evening Shift

Light from a point source in air falls on a spherical glass surface (refractive index, $\mu = 1.5$ and radius of curvature $= 50\ \mathrm{cm}$). The image is formed at a distance of 200 cm from the glass surface inside the glass. The magnitude of distance of the light source from the glass surface is _____m.

Open solution

Q228. JEE Main 2025 (Online) 2nd April Evening Shift

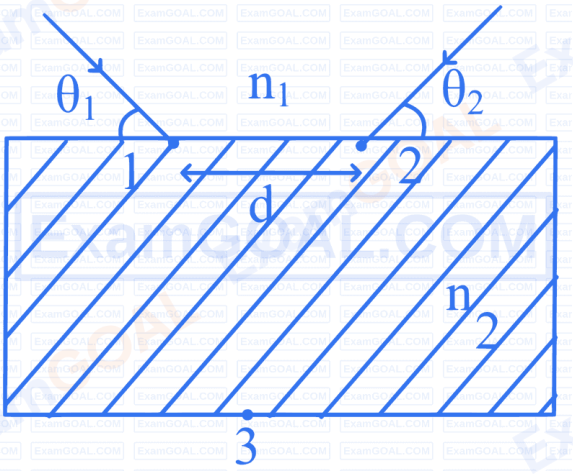
A ray of light suffers minimum deviation when incident on a prism having angle of the prism equal to 60°. The refractive index of the prism material is $\sqrt{2}$. The angle of incidence (in degrees) is_____

Open solution

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Q229. JEE Main 2025 (Online) 29th January Morning Shift

Two light beams fall on a transparent material block at point 1 and 2 with angle θ_1 and θ_2, respectively, as shown in figure. After refraction, the beams intersect at point 3 which is exactly on the interface at other end of the block. Given : the distance between 1 and 2, $\mathrm{d} = 4\sqrt{3}\ \mathrm{cm}$ and $\theta_1 = \theta_2 = \cos^{-1}\left(\frac{n_2}{2n_1}\right)$. where refractive index of the block $n_2 >$ refractive index of the outside medium n_1, then the thickness of the block is _____ cm .



Open solution

Q230. JEE Main 2025 (Online) 22nd January Morning Shift

The driver sitting inside a parked car is watching vehicles approaching from behind with the help of his side view mirror, which is a convex mirror with radius of curvature $\mathrm{R} = 2\ \mathrm{m}$. Another car approaches him from behind with a uniform speed of $90\ \mathrm{km/hr}$. When the car is at a distance of 24 m from him, the magnitude of the acceleration of the image of the car in the side view mirror is ' a '. The value of $100a$ is _____ $\mathrm{m/s^2}$.

Open solution

Q231. JEE Main 2024 (Online) 6th April Morning Shift

The refractive index of prism is $\mu = \sqrt{3}$ and the ratio of the angle of minimum deviation to the angle of prism is one. The value of angle of prism is _____°.

Open solution

Q232. JEE Main 2024 (Online) 4th April Evening Shift

A light ray is incident on a glass slab of thickness $4\sqrt{3}\ \mathrm{cm}$ and refractive index $\sqrt{2}$ The angle of incidence is equal to the critical angle for the glass slab with air. The lateral displacement of ray after passing through glass slab is _____ cm.

(Given $\sin 15^\circ = 0.25$)

Open solution

Q233. JEE Main 2024 (Online) 1st February Morning Shift

The distance between object and its 3 times magnified virtual image as produced by a convex lens is $20\ \mathrm{cm}$. The focal length of the lens used is _____ cm.

Open solution

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Q234. JEE Main 2024 (Online) 31st January Evening Shift

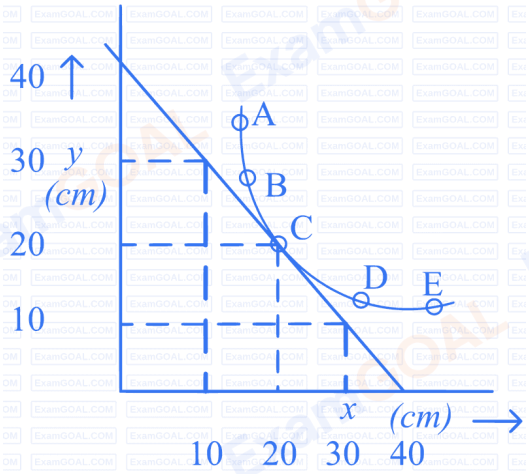
Light from a point source in air falls on a convex curved surface of radius $20\ \text{cm}$ and refractive index 1.5. If the source is located at $100\ \text{cm}$ from the convex surface, the image will be formed at _____ cm from the object.

Open solution

Q235. JEE Main 2024 (Online) 30th January Evening Shift

In an experiment to measure the focal length (f) of a convex lens, the magnitude of object distance (x) and the image distance (y) are measured with reference to the focal point of the lens. The y-x plot is shown in figure.

The focal length of the lens is _____ cm.



Open solution

Q236. JEE Main 2024 (Online) 30th January Morning Shift

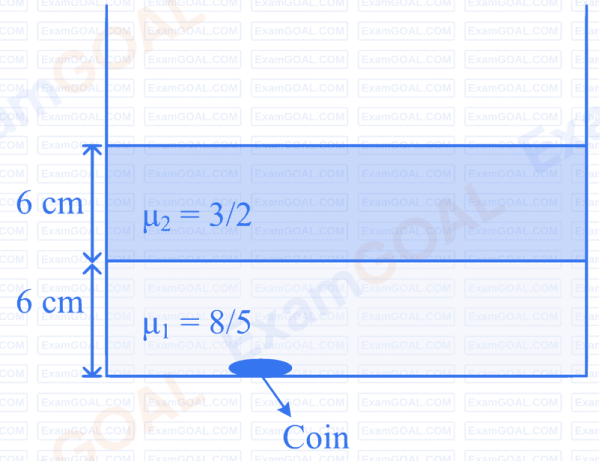
The distance between object and its two times magnified real image as produced by a convex lens is $45\ \text{cm}$. The focal length of the lens used is _____ cm.

Open solution

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Q237. JEE Main 2024 (Online) 27th January Morning Shift

Two immiscible liquids of refractive indices $\frac{8}{5}$ and $\frac{3}{2}$ respectively are put in a beaker as shown in the figure. The height of each column is $6\ \text{cm}$. A coin is placed at the bottom of the beaker. For near normal vision, the apparent depth of the coin is $\frac{\alpha}{4}\ \text{cm}$. The value of α is _____.



Open solution

Q238. JEE Main 2023 (Online) 15th April Morning Shift

The refractive index of a transparent liquid filled in an equilateral hollow prism is $\sqrt{2}$. The angle of minimum deviation for the liquid will be _____ $^\circ$.

Open solution

Q239. JEE Main 2023 (Online) 13th April Evening Shift

A bi convex lens of focal length $10\ \text{cm}$ is cut in two identical parts along a plane perpendicular to the principal axis. The power of each lens after cut is _____ D.

Open solution

Q240. JEE Main 2023 (Online) 13th April Morning Shift

A fish rising vertically upward with a uniform velocity of $8\ \text{ms}^{-1}$, observes that a bird is diving vertically downward towards the fish with the velocity of $12\ \text{ms}^{-1}$. If the refractive index of water is $\frac{4}{3}$, then the actual velocity of the diving bird to pick the fish, will be _____ ms^{-1}.

Open solution

Q241. JEE Main 2023 (Online) 12th April Morning Shift

Two convex lenses of focal length $20\ \text{cm}$ each are placed coaxially with a separation of $60\ \text{cm}$ between them. The image of the distant object formed by the combination is at _____ cm from the first lens.

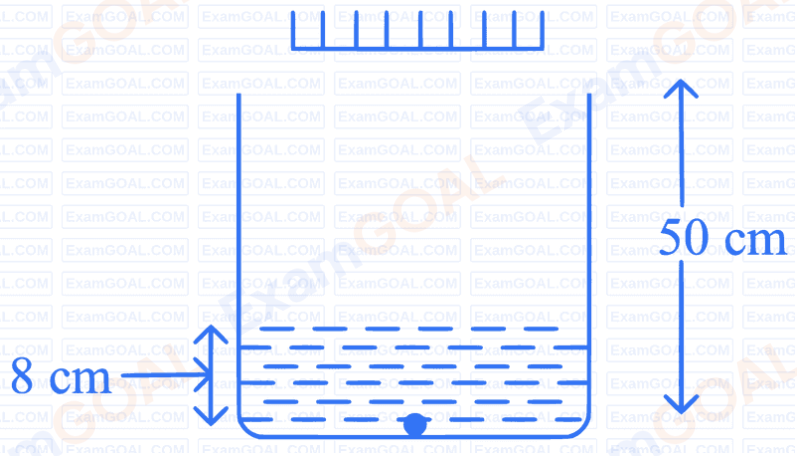
Open solution

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Q242. JEE Main 2023 (Online) 11th April Evening Shift

Open solution

As shown in the figure, a plane mirror is fixed at a height of $50\ \text{cm}$ from the bottom of tank containing water $\left(\mu = \frac{4}{3}\right)$. The height of water in the tank is $8\ \text{cm}$. A small bulb is placed at the bottom of the water tank. The distance of image of the bulb formed by mirror from the bottom of the tank is _____ cm.



Q243. JEE Main 2023 (Online) 11th April Morning Shift

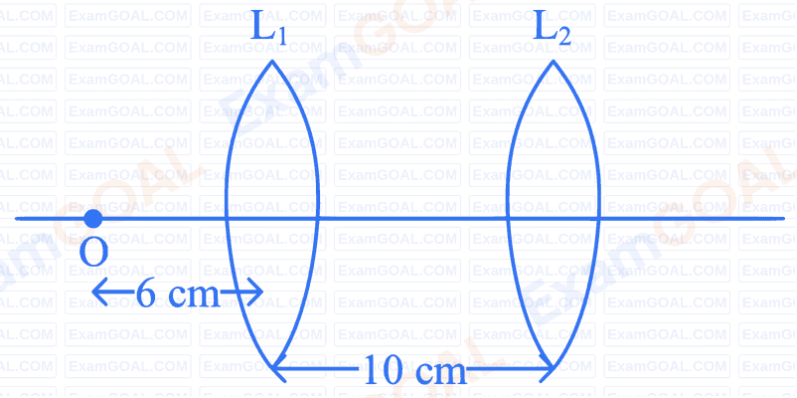
Open solution

The radius of curvature of each surface of a convex lens having refractive index 1.8 is $20\ \text{cm}$. The lens is now immersed in a liquid of refractive index 1.5 . The ratio of power of lens in air to its power in the liquid will be $x : 1$. The value of x is _____.

Q244. JEE Main 2023 (Online) 10th April Evening Shift

Open solution

A point object, 'O' is placed in front of two thin symmetrical coaxial convex lenses L_1 and L_2 with focal length $24\ \text{cm}$ and $9\ \text{cm}$ respectively. The distance between two lenses is $10\ \text{cm}$ and the object is placed $6\ \text{cm}$ away from lens L_1 as shown in the figure. The distance between the object and the image formed by the system of two lenses is _____ cm.



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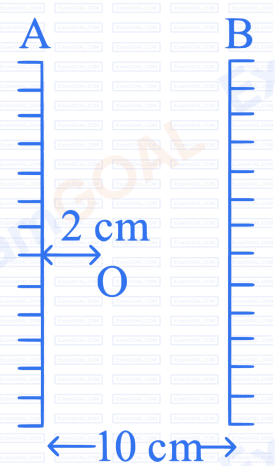
Q245. JEE Main 2023 (Online) 8th April Evening Shift

Two transparent media having refractive indices 1.0 and 1.5 are separated by a spherical refracting surface of radius of curvature $30\ \text{cm}$. The centre of curvature of surface is towards denser medium and a point object is placed on the principal axis in rarer medium at a distance of $15\ \text{cm}$ from the pole of the surface. The distance of image from the pole of the surface is _____ cm.

Open solution

Q246. JEE Main 2023 (Online) 8th April Morning Shift

Two vertical parallel mirrors A and B are separated by $10\ \text{cm}$. A point object O is placed at a distance of $2\ \text{cm}$ from mirror A. The distance of the second nearest image behind mirror A from the mirror A is _____ cm.



Open solution

Q247. JEE Main 2023 (Online) 6th April Morning Shift

A pole is vertically submerged in swimming pool, such that it gives a length of shadow $2.15\ \text{m}$ within water when sunlight is incident at angle of 30° with the surface of water. If swimming pool is filled to a height of $1.5\ \text{m}$, then the height of the pole above the water surface in centimeters is $(n_w = 4/3)$ _____.

Open solution

Q248. JEE Main 2023 (Online) 1st February Morning Shift

A thin cylindrical rod of length $10\ \text{cm}$ is placed horizontally on the principal axis of a concave mirror of focal length $20\ \text{cm}$. The rod is placed in such a way that mid point of the rod is at $40\ \text{cm}$ from the pole of mirror. The length of the image formed by the mirror will be $\frac{x}{3}\ \text{cm}$. The value of x is _____.

Open solution

Q249. JEE Main 2023 (Online) 30th January Morning Shift

In an experiment for estimating the value of focal length of converging mirror, image of an object placed at $40\ \text{cm}$ from the pole of the mirror is formed at distance $120\ \text{cm}$ from the pole of the mirror. These distances are measured with a modified scale in which there are 20 small divisions in $1\ \text{cm}$. The value of error in measurement of focal length of the mirror is $\frac{1}{\text{K}}\ \text{cm}$. The value of K is _____.

Open solution

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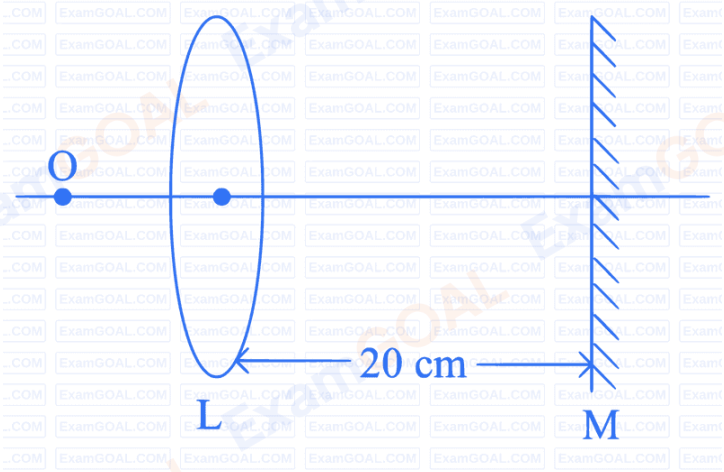
Q250. JEE Main 2023 (Online) 29th January Evening Shift

In an experiment of measuring the refractive index of a glass slab using travelling microscope in physics lab, a student measures real thickness of the glass slab as 5.25 mm and apparent thickness of the glass slab as 5.00 mm. Travelling microscope has 20 divisions in one cm on main scale and 50 divisions on vernier scale is equal to 49 divisions on main scale. The estimated uncertainty in the measurement of refractive index of the slab is $\frac{x}{10} \times 10^{-3}$, where x is _____

Open solution

Q251. JEE Main 2023 (Online) 25th January Evening Shift

An object is placed on the principal axis of convex lens of focal length 10cm as shown. A plane mirror is placed on the other side of lens at a distance of 20 cm. The image produced by the plane mirror is 5cm inside the mirror. The distance of the object from the lens is _____ cm.



Open solution

Q252. JEE Main 2023 (Online) 25th January Morning Shift

A ray of light is incident from air on a glass plate having thickness $\sqrt{3}$ cm and refractive index $\sqrt{2}$. The angle of incidence of a ray is equal to the critical angle for glass-air interface. The lateral displacement of the ray when it passes through the plate is _____ $\times 10^{-2}$ cm. (given $\sin 15^\circ = 0.26$)

Open solution

Q253. JEE Main 2023 (Online) 24th January Evening Shift

A convex lens of refractive index 1.5 and focal length 18cm in air is immersed in water. The change in focal length of the lens will be _____ cm.

(Given refractive index of water $= \frac{4}{3}$)

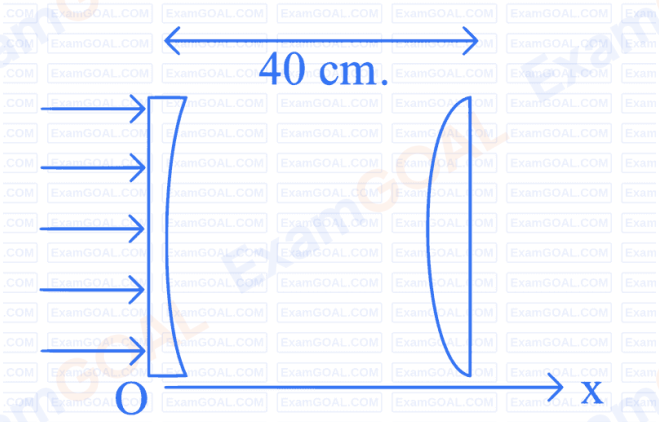
Open solution

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Q254. JEE Main 2023 (Online) 24th January Morning Shift

Open solution

As shown in the figure, a combination of a thin plano concave lens and a thin plano convex lens is used to image an object placed at infinity. The radius of curvature of both the lenses is 30 cm and refraction index of the material for both the lenses is 1.75. Both the lenses are placed at distance of 40 cm from each other. Due to the combination, the image of the object is formed at distance $x =$ _____ cm, from concave lens.



Q255. JEE Main 2022 (Online) 30th July Evening Shift

Open solution

A ring is placed at the bottom of a trough containing four immiscible liquids of refractive indices 1.0, 2.0, 3.0 and 4.0 poured one above the other of heights $10\ \text{cm}$, $20\ \text{cm}$, $30\ \text{cm}$, and $40\ \text{cm}$ respectively. The apparent depth at which the ring appears, when seen from outside is _____ cm.

Q256. JEE Main 2022 (Online) 29th July Morning Shift

Open solution

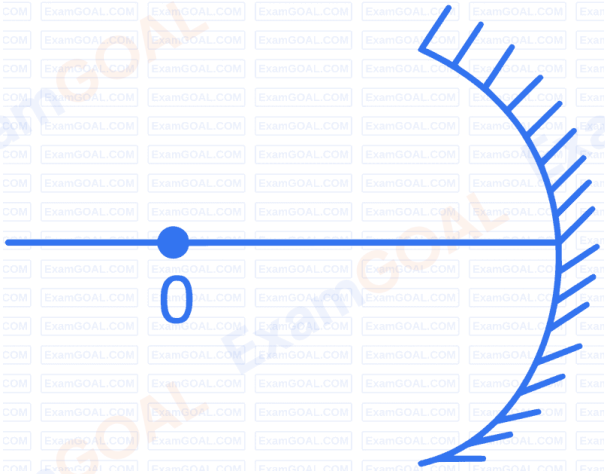
The X-Y plane be taken as the boundary between two transparent media M_1 and M_2. M_1 in $Z \geqslant 0$ has a refractive index of $\sqrt{2}$ and M_2 with $Z < 0$ has a refractive index of $\sqrt{3}$. A ray of light travelling in M_1 along the direction given by the vector $\overrightarrow{\mathrm{P}} = 4\sqrt{3}\hat{i} - 3\sqrt{3}\hat{j} - 5\hat{k}$, is incident on the plane of separation. The value of difference between the angle of incidence in M_1 and the angle of refraction in M_2 will be _____ degree.

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Q257. JEE Main 2022 (Online) 28th July Evening Shift

An object 'O' is placed at a distance of $100\ \text{cm}$ in front of a concave mirror of radius of curvature $200\ \text{cm}$ as shown in the figure. The object starts moving towards the mirror at a speed $2\ \text{cm/s}$. The position of the image from the mirror after $10\ \text{s}$ will be at _____ cm.

Open solution



Q258. JEE Main 2022 (Online) 28th July Evening Shift

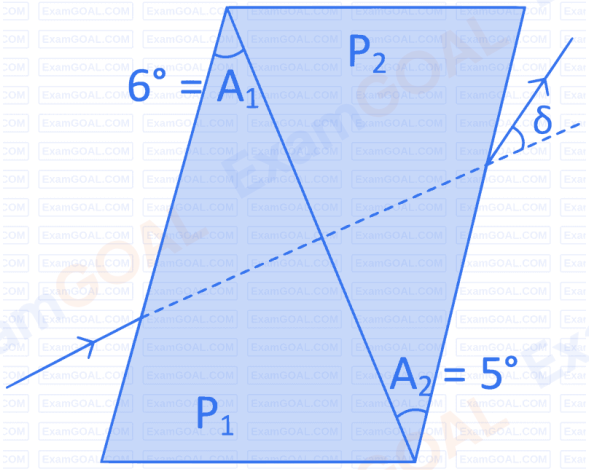
In an experiment with a convex lens, The plot of the image distance (v') against the object distance (μ') measured from the focus gives a curve $v'\mu' = 225$. If all the distances are measured in cm. The magnitude of the focal length of the lens is _____ cm.

Open solution

Q259. JEE Main 2022 (Online) 27th July Evening Shift

A thin prism of angle 6° and refractive index for yellow light $(\text{n}_\text{Y})1.5$ is combined with another prism of angle 5° and $\text{n}_\text{Y} = 1.55$. The combination produces no dispersion. The net average deviation (δ) produced by the combination is $\left(\frac{1}{x}\right)^\circ$. The value of x is _____.

Open solution

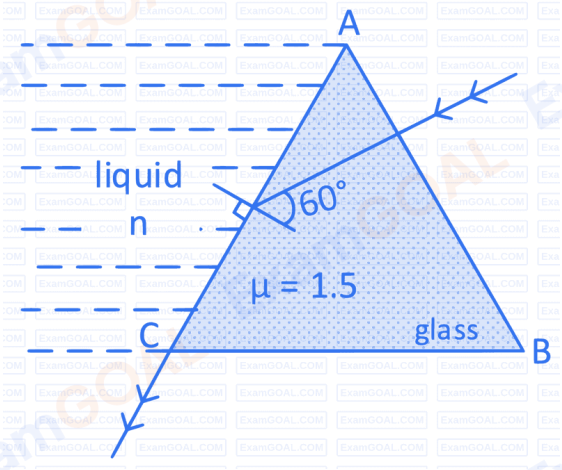


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Q260. JEE Main 2022 (Online) 26th July Evening Shift

In the given figure, the face AC of the equilateral prism is immersed in a liquid of refractive index 'n'. For incident angle 60° at the side AC, the refracted light beam just grazes along face AC. The refractive index of the liquid $n = \frac{\sqrt{x}}{4}$. The value of x is _____.

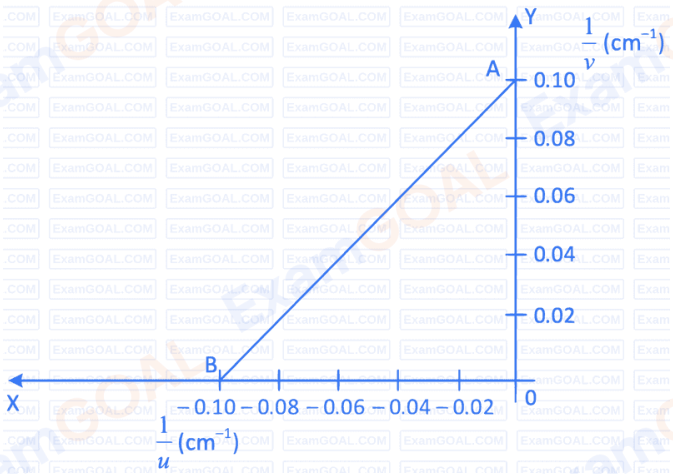
(Given refractive index of glass $= 1.5$)



Open solution

Q261. JEE Main 2022 (Online) 26th July Morning Shift

The graph between $\frac{1}{u}$ and $\frac{1}{v}$ for a thin convex lens in order to determine its focal length is plotted as shown in the figure. The refractive index of lens is 1.5 and its both the surfaces have same radius of curvature R. The value of R will be _____ cm. (where $u =$ object distance, $v =$ image distance)



Open solution

Q262. JEE Main 2022 (Online) 25th July Evening Shift

A convex lens of focal length 20 cm is placed in front of a convex mirror with principal axis coinciding each other. The distance between the lens and mirror is 10 cm. A point object is placed on principal axis at a distance of 60 cm from the convex lens. The image formed by combination coincides the object itself. The focal length of the convex mirror is _____ cm.

Open solution

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Q263. JEE Main 2022 (Online) 30th June Morning Shift

The refractive index of an equilateral prism is $\sqrt{2}$. The angle of emergence under minimum deviation position of prism, in degree, is _____.

Open solution

Q264. JEE Main 2022 (Online) 29th June Morning Shift

A parallel beam of light is allowed to fall on a transparent spherical globe of diameter 30 cm and refractive index 1.5. The distance from the centre of the globe at which the beam of light can converge is _____ mm.

Open solution

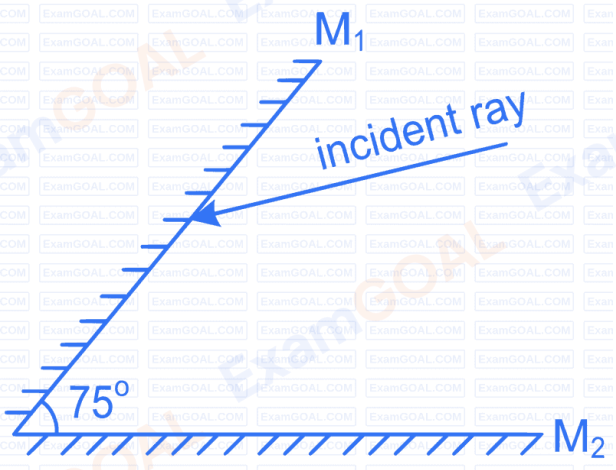
Q265. JEE Main 2022 (Online) 26th June Evening Shift

A small bulb is placed at the bottom of a tank containing water to a depth of $\sqrt{7}$ m. The refractive index of water is $\frac{4}{3}$. The area of the surface of water through which light from the bulb can emerge out is $x\pi$ m^2. The value of x is _____.

Open solution

Q266. JEE Main 2022 (Online) 26th June Morning Shift

A light ray is incident, at an incident angle θ_1, on the system of two plane mirrors M_1 and M_2 having an inclination angle 75° between them (as shown in figure). After reflecting from mirror M_1 it gets reflected back by the mirror M_2 with an angle of reflection 30°. The total deviation of the ray will be _____ degree.



Open solution

Q267. JEE Main 2022 (Online) 24th June Evening Shift

A ray of light is incident at an angle of incidence 60° on the glass slab of refractive index $\sqrt{3}$. After refraction, the light ray emerges out from other parallel faces and lateral shift between incident ray and emergent ray is $4\sqrt{3}$ cm. The thickness of the glass slab is _____ cm.

Open solution

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Q268. JEE Main 2022 (Online) 24th June Morning Shift

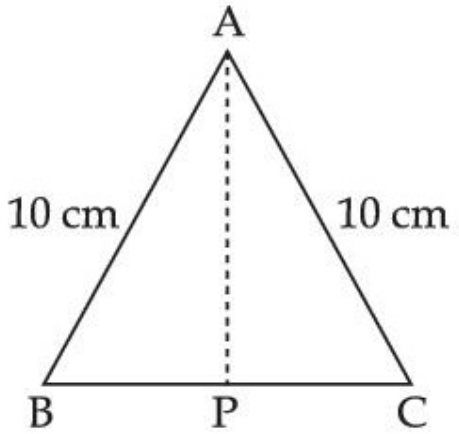
Two identical thin biconvex lens of focal length 15 cm and refractive index 1.5 are in contact with each other. The space between the lenses is filled with a liquid of refractive index 1.25. The focal length of the combination is _____ cm.

Open solution

Q269. JEE Main 2021 (Online) 31st August Evening Shift

Cross-section view of a prism is the equilateral triangle ABC in the figure. The minimum deviation is observed using this prism when the angle of incidence is equal to the prism angle. The time taken by light to travel from P (midpoint of BC) to A is _____ $\times 10^{-10}$ s. (Given, speed of light in vacuum = 3×10^8 m/s and $\cos 30^\circ = \frac{\sqrt{3}}{2}$)

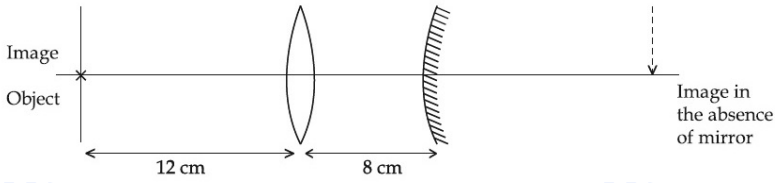
Open solution



Q270. JEE Main 2021 (Online) 26th August Evening Shift

An object is placed at a distance of 12 cm from a convex lens. A convex mirror of focal length 15 cm is placed on other side of lens at 8 cm as shown in the figure. Image of object coincides with the object.

Open solution



When the convex mirror is removed, a real and inverted image is formed at a position. The distance of the image from the object will be ..... (cm)

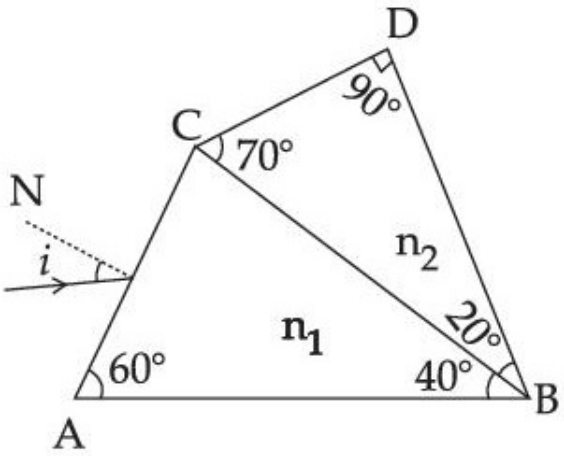
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Q271. JEE Main 2021 (Online) 27th July Morning Shift

A prism of refractive index n_1 and another prism of refractive index n_2 are stuck together (as shown in the figure). n_1 and n_2 depend on λ, the wavelength of light, according to the relation

$n_1 = 1.2 + \frac{10.8\times10^{-14}}{\lambda^2}$ and $n_2 = 1.45 + \frac{1.8\times10^{-14}}{\lambda^2}$

The wavelength for which rays incident at any angle on the interface BC pass through without bending at that interface will be _____ nm.



Open solution

Q272. JEE Main 2021 (Online) 22th July Evening Shift

A ray of light passing through a prism ($\mu = \sqrt{3}$) suffers minimum deviation. It is found that the angle of incidence is double the angle of refraction within the prism. Then, the angle of prism is _____ (in degrees).

Open solution

Q273. JEE Main 2021 (Online) 20th July Morning Shift

An object viewed from a near point distance of 25 cm, using a microscopic lens with magnification '6', gives an unresolved image. A resolved image is observed at infinite distance with a total magnification double the earlier using an eyepiece along with the given lens and a tube of length 0.6 m, if the focal length of the eyepiece is equal to _____ cm.

Open solution

Q274. JEE Main 2021 (Online) 17th March Evening Shift

The image of an object placed in air formed by a convex refracting surface is at a distance of 10 m behind the surface. The image is real and is at $\frac{2}{3}^{rd}$ of the distance of the object from the surface. The wavelength of light inside the surface is $\frac{2}{3}$ times the wavelength in air. The radius of the curved surface is $\frac{x}{13}$ m. The value of 'x' is _____.

Open solution

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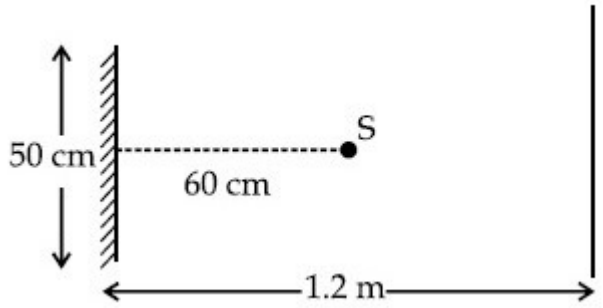
Q275. JEE Main 2021 (Online) 16th March Evening Shift

A deviation of 2° is produced in the yellow ray when prism of crown and flint glass are achromatically combined. Taking dispersive powers of crown and flint glass as 0.02 and 0.03 respectively and refractive index for yellow light for these glasses are 1.5 and 1.6 respectively. The refracting angles for crown glass prism will be _____° (in degree). (Round off to the Nearest Integer)

Open solution

Q276. JEE Main 2021 (Online) 26th February Evening Shift

A point source of light S, placed at a distance 60cm in front of the centre of a plane mirror of width 50 cm, hangs vertically on a wall. A man walks in front of the mirror along a line parallel to the mirror at a distance 1.2 m from it (see in the figure). The distance between the extreme points where he can see the image of the light source in the mirror is _____ cm.



Open solution

Q277. JEE Main 2021 (Online) 25th February Morning Shift

The same size images are formed by a convex lens when the object is placed at 20 cm or at 10 cm from the lens. The focal length of convex lens is _____ cm.

Open solution

Q278. JEE Main 2020 (Online) 5th September Evening Slot

A prism of angle A = 1° has a refractive index μ = 1.5. A good estimate for the minimum angle of deviation (in degrees) is close to $\frac{N}{10}$.
Value of N is ___.

Open solution

Q279. JEE Main 2020 (Online) 5th September Morning Slot

A compound microscope consists of an objective lens of focal length 1 cm and an eyepiece of focal length 5 cm with a separation of 10 cm.
The distance between an object and the objective lens, at which the strain on the eye is minimum is $\frac{n}{40}$ cm. The value of n is ____.

Open solution

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Q280. JEE Main 2020 (Online) 4th September Evening Slot

The distance between an object and a screen is 100 cm. A lens can produce real image of the object on the screen for two different positions between the screen and the object. The distance between these two positions is 40 cm. If the power of the lens is close to $\left(\frac{N}{100}\right)D$ where N is an integer, the value of N is _____.

Open solution

Q281. JEE Main 2020 (Online) 4th September Morning Slot

In a compound microscope, the magnified virtual image is formed at a distance of 25 cm from the eye-piece. The focal length of its objective lens is 1 cm. If the magnification is 100 and the tube length of the microscope is 20 cm, then the focal length of the eye-piece lens (in cm) is _____.

Open solution

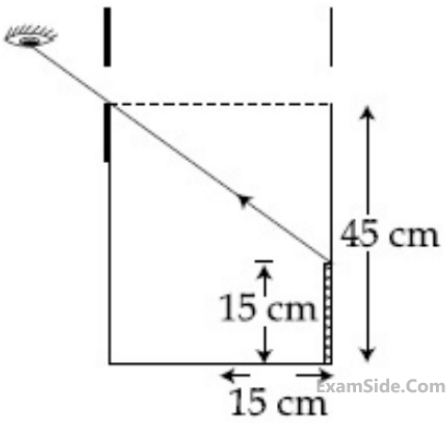
Q282. JEE Main 2020 (Online) 3rd September Evening Slot

When an object is kept at a distance of 30 cm from a concave mirror, the image is formed at a distance of 10 cm from the mirror. If the object is moved with a speed of 9 cms^{-1}, the speed (in cms^{-1}) with which image moves at that instant is ____.

Open solution

Q283. JEE Main 2020 (Online) 3rd September Morning Slot

An observer can see through a small hole on the side of a jar (radius 15 cm) at a point at height of 15 cm from the bottom (see figure). The hole is at a height of 45 cm. When the jar is filled with a liquid up to a height of 30 cm the same observer can see the edge at the bottom of the jar. If the refractive index of the liquid is N/100, where N is an integer, the value of N is _____.



Open solution

Q284. JEE Main 2020 (Online) 2nd September Evening Slot

A light ray enters a solid glass sphere of refractive index $\mu = \sqrt{3}$ at an angle of incidence 60°. The ray is both reflected and refracted at the farther surface of the sphere. The angle (in degrees) between the reflected and refracted rays at this surface is _____.

Open solution

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Q285. JEE Main 2020 (Online) 8th January Morning Slot

A point object in air is in front of the curved surface of a plano-convex lens. The radius of curvature of the curved surface is 30 cm and the refractive index of the lens material is 1.5, then the focal length of the lens (in cm) is _____.

Open solution

MCQ (More than One Correct Answer) 1 questions

Q286. JEE Main 2026 (Online) 23rd January Evening Shift

A prism of angle 75° and refractive index $\sqrt{3}$ is coated with thin film of refractive index 1.5 only at the back exit surface. To have total internal reflection at the back exit surface the incident angle must be ____

$(\sin 15^\circ = 0.25$ and $\sin 25^\circ = 0.43)$

(A) between 15° and 20°

(B) 15°

(C) $< 15^\circ$

(D) $> 25^\circ$

Open solution

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Answer Key

MCQ (Single Correct Answer)

1. **A**	2. **A**	3. **C**	4. **A**	5. **A**
6. **B**	7. **B**	8. **A**	9. **D**	10. **B**
11. **D**	12. **B**	13. **A**	14. **A**	15. **B**
16. **B**	17. **D**	18. **C**	19. **A**	20. **B**
21. **C**	22. **B**	23. **D**	24. **C**	25. **B**
26. **D**	27. **C**	28. **B**	29. **B**	30. **B**
31. **C**	32. **B**	33. **D**	34. **C**	35. **D**
36. **D**	37. **D**	38. **C**	39. **B**	40. **B**
41. **C**	42. **C**	43. **A**	44. **B**	45. **A**
46. **D**	47. *Scan QR*	48. **B**	49. **A**	50. **C**
51. **D**	52. **D**	53. **A**	54. **B**	55. **B**
56. **B**	57. **A**	58. **D**	59. **A**	60. **D**
61. **A**	62. **A**	63. **B**	64. **A**	65. **A**
66. **B**	67. **D**	68. **B**	69. **A**	70. **B**
71. **C**	72. **A**	73. **B**	74. **D**	75. **A**
76. **D**	77. **B**	78. **A**	79. **C**	80. **A**
81. **D**	82. **D**	83. **D**	84. **A**	85. **D**
86. **C**	87. **C**	88. **A**	89. **A**	90. **A**
91. **D**	92. **C**	93. **B**	94. **A**	95. **A**
96. **B**	97. **A**	98. **B**	99. **A**	100. **A**
101. **D**	102. **A**	103. **A**	104. **D**	105. **D**
106. **C**	107. **B**	108. **C**	109. **B**	110. **B**
111. **A**	112. **A**	113. **D**	114. **B**	115. **A**
116. **A**	117. **C**	118. **A**	119. **D**	120. **D**
121. **B**	122. **B**	123. **B**	124. **B**	125. **C**
126. **B**	127. **C**	128. **B**	129. **C**	130. **A**
131. **A**	132. **D**	133. **A**	134. **A**	135. **C**
136. **C**	137. **B**	138. **C**	139. **C**	140. **A**
141. **B**	142. **D**	143. **D**	144. **C**	145. **C**
146. **D**	147. **A**	148. **B**	149. **B**	150. **C**

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151. **A**	152. **C**	153. **A**	154. **C**	155. **B**
156. **A**	157. **A**	158. **A**	159. **B**	160. **C**
161. **B**	162. **D**	163. **B**	164. **D**	165. **B**
166. **B**	167. **A**	168. **D**	169. **B**	170. **B**
171. **C**	172. **B**	173. **A**	174. **B**	175. **C**
176. **B**	177. **C**	178. **D**	179. **C**	180. **A**
181. **D**	182. **A**	183. **D**	184. **D**	185. **B**
186. **B**	187. *Scan QR*	188. **C**	189. **D**	190. **D**
191. **D**	192. **B**	193. **C**	194. **C**	195. **D**
196. **C**	197. **C**	198. **A**	199. **D**	200. **A**
201. **A**	202. **B**	203. **A**	204. **D**	205. **C**
206. **C**	207. **C**	208. **A**	209. **A**	210. **B**
211. **C**	212. **B**	213. **C**	214. **B**	215. **B**
216. **A**	217. **A**			

Numerical

218. **10**	219. **20**	220. **3**	221. **16**	222. **100**
223. **100**	224. **280**	225. **80**	226. **45**	227. **4**
228. **45**	229. **6**	230. **8**	231. **60**	232. **2**
233. **15**	234. **200**	235. **20**	236. **10**	237. **31**
238. **30**	239. **5**	240. **3**	241. **100**	242. **98**
243. **4**	244. **34**	245. **30**	246. **18**	247. **50**
248. **32**	249. **32**	250. **41**	251. **30**	252. **52**
253. **54**	254. **120**	255. **40**	256. **15**	257. **400**
258. **15**	259. **4**	260. **27**	261. **10**	262. **10**
263. **45**	264. **225**	265. **9**	266. **210**	267. **12**
268. **10**	269. **5**	270. **50**	271. **600**	272. **60**
273. **25**	274. **30**	275. **12**	276. **150**	277. **15**
278. **5**	279. **50**	280. **476**	281. **6.25**	282. **1**
283. **158**	284. **90**	285. **60**		

MCQ (More than One Correct Answer)

286. **A, B, C**	

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For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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